Smart and Sustainable City of Seoul – Gangdong-gu

Energy Management, Phase 2 Report

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KTH Royal Institute of Technology, Sustainable Energy Engineering

Roles and Responsibilities

This project is being conducted as part of the Sustainable Energy Engineering Master's program at KTH Royal Institute of Technology. In collaboration with the city of Seoul, KTH is supporting an innovative initiative focused on designing the future energy landscape of Gangdong-gu as part of the Energy Management course.

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Executive Summary

This report is conducted by a team of students from the *Energy Management* course (MJ2410) at KTH Royal Institute of Technology, Stockholm, in collaboration with the city of Seoul. The project focuses on designing a smart and sustainable energy future for Gangdong-gu, a district of approximately 500,000 residents in eastern Seoul, structured around three Key Performance Indicators: nearly zero energy, nearly zero emissions, and peak shaving.

Gangdong-gu's total final energy consumption amounts to approximately 4,340 GWh per year, with primary energy demand reaching 7,500 GWh — a gap driven by fossil-fuel dependency and the dominance of gas boilers in building heating. In 2020, the district emitted approximately 1,359 ktCO_2 eq, of which 71% originated from the building sector.

Three scenarios of increasing ambition were developed and modelled using nPro. Business as Usual (BAU) reflects a conservative transition aligned with South Korea's Nationally Determined Contribution. District Energy Consideration (DEC) introduces 5th Generation District Heating and Cooling (5GDHC), expanded heat pump deployment, battery storage, and coordinated EV charging. Grid-Linked Decarbonization (GLD) represents the most ambitious pathway, fully phasing out gas boilers and deploying additional renewable capacity beyond the district boundary.

GLD achieves the strongest environmental performance, reducing CO_2 emissions by 62% and primary energy demand by 45% relative to 2025. DEC delivers a 57% emissions reduction and 42% primary energy reduction, while BAU reduces emissions by 40% and primary energy by 28%. All scenarios are financially viable; DEC demonstrates the strongest business case with an NPV of 1,042 million USD, an IRR of 12.2%, and a discounted payback period of 9 years.

The DEC scenario is recommended as the primary implementation pathway, combining strong environmental performance, superior economic viability, and practical feasibility, supported by the proposed *Gangdong Energy Hub* public-private Energy-as-a-Service business model. Although GLD achieves the strongest environmental performance, its higher investment requirement and longer payback period reduce its practical attractiveness compared with DEC.

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1 Introduction

Gangdong-gu, located in eastern Seoul, has approximately 500,000 residents [42] and covers 24.58 km^2. Land use is 57.21% residential, 40.32% green space, and 2.47% commercial [14]. While total population remained stable between 2015 and 2020, residential expansion shifted westward and outward [8]. Ongoing large-scale redevelopment of ageing apartment complexes [41] has increased the population by about 50,000 over the past five years, raising housing density and energy demand [8, 42]. Final energy use reached 4,333 GWh in 2023 [24], dominated by the residential sector. In terms of industry Gangdong-gu has two types of industry, one wastewater treatment plant and one train depot.

Gangdong-gu experiences hot and humid summers and cold winters, which creates both significant cooling demand in summer and heating demand in winter. The district is dominated by dense residential apartment developments, supported by schools, public facilities, retail areas, and small commercial activities. Daily mobility is strongly influenced by commuting patterns within the Seoul metropolitan area, making transport electrification and charging coordination relevant for future peak-load management.

This report provides execution-ready recommendations for a smart and sustainable district by 2035, supported by a full urban energy system analysis for 2035–2050. The district profile shows that Gangdong-gu's energy transition challenge is primarily a dense urban building-energy problem, complemented by increasing electricity demand from cooling and future transport electrification. Therefore, the scenario design focuses on building efficiency, heating electrification, district energy integration, renewable electricity, storage, and coordinated EV charging.

Table 1: District profile of Gangdong-gu

Category	**Key data**	**Relevance for the project**
Location and area	Eastern Seoul; 24.58 km^2	Dense urban district within the Seoul metropolitan energy and mobility system.
Population	Approx. 500,000 residents	High residential energy demand and strong relevance of building-related measures.
Land use	57.21% residential, 40.32% green space, 2.47% commercial	Residential buildings dominate the energy transition potential.
Urban structure	Dense apartment developments, schools, public facilities, retail areas	Suitable for coordinated district energy, retrofitting, PV, storage, and smart charging.
Climate	Hot, humid summers and cold winters	Creates both significant cooling demand and strong heating demand.
Economic and industrial activities	Mainly residential and service-oriented; wastewater treatment plant and train depot	Limited industrial emissions; focus is on buildings, mobility, and public infrastructure.
Current final energy use	Approx. 4,333–4,340 GWh/year	Energy demand is dominated by building-related consumption.
Main energy challenge	Imported energy dependence, gas boilers, carbon-intensive electricity	Requires electrification, efficiency improvements, renewable energy, and storage.
Main emission challenge	Building energy use dominates district emissions	Heating decarbonisation and electricity decarbonisation are key mitigation priorities.

1.1 Problem framing and KPI motivation

Gangdong-gu faces three interlinked energy system challenges that motivate the KPI framework used in this report. First, the district remains strongly dependent on imported energy and fossil-fuel-based supply, particularly through natural gas heating and electricity generated from a carbon-intensive national grid. This creates the need for the first KPI, nearly zero energy, which focuses on reducing net external energy dependence through efficiency improvements, electrification, renewable generation, and energy recovery.

Second, greenhouse gas emissions are dominated by the building sector, especially space heating and electricity demand in dense residential apartment areas. This motivates the second KPI, nearly zero emissions, which evaluates whether the proposed scenarios can substantially reduce the remaining carbon footprint of buildings, mobility, waste-related activities, and imported electricity.

Third, electrification of heating and transport can reduce fossil fuel use but may also increase stress on the electricity grid. Higher heat pump deployment and EV charging can create new peak-load events if they are not coordinated. This motivates the third KPI, peak shaving, which assesses whether storage, smart charging, demand management, and district energy integration can reduce maximum grid import and support a more stable electricity system.

The three KPIs are therefore not independent targets, but connected responses to the main transition challenges of Gangdong-gu: reducing energy dependence, lowering emissions, and avoiding new electricity-grid constraints during electrification.

1.2 Energy Supply and Demand

Figure 1 shows the final energy mix used for Seoul's energy supply by energy source. Based on the 2023 energy balance, the final energy supply remains strongly dependent on fossil fuels, while renewable energy sources account for a comparatively small share of total final energy consumption [35, 37]. This final energy mix forms the basis for the subsequent primary-to-final energy flow analysis.

Figure 3 illustrates the energy flow from primary energy (PE) supply to final energy (FE) consumption and useful energy services based on the 2023 energy balance [35, 37]. The analysis explicitly accounts for conversion efficiencies, sectoral end-use shares, and

system losses, complemented with international benchmark data [12, 17, 20].

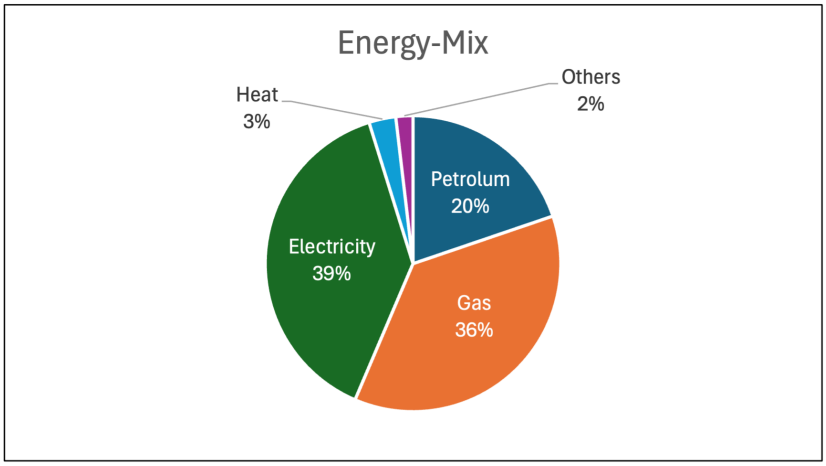

Figure 1: Final energy consumption mix in Seoul by energy source.

Total final energy consumption in the district amounts to approximately 4,340 GWh per year, while the corresponding primary energy demand reaches about 7,500 GWh [35]. Conversion losses in electricity generation, fuel transformation, and transport processes account for roughly 4,184 GWh, highlighting the inefficiency of the current system. Electricity is assigned a primary energy factor (PEF) of 2.75, reflecting a fossil-dominated generation mix, while direct fossil fuel use is assigned a PEF of 1.2 [12].

The electricity supply of Seoul is dominated by fossil fuels and nuclear power. According to official statistics from the Seoul Energy Information System and national electricity balance data, coal accounts for approximately 31.4% of electricity generation, nuclear power for 30.7%, and liquefied natural gas (LNG) for 26.8% [20, 37]. Renewable energy sources contribute only about 8.4% to the electricity mix, while oil and other sources together account for less than 3% [37]. This generation structure, illustrated in Figure 2, explains the relatively high primary energy factor of electricity used in the analysis and the substantial upstream conversion losses observed in Figure 3.

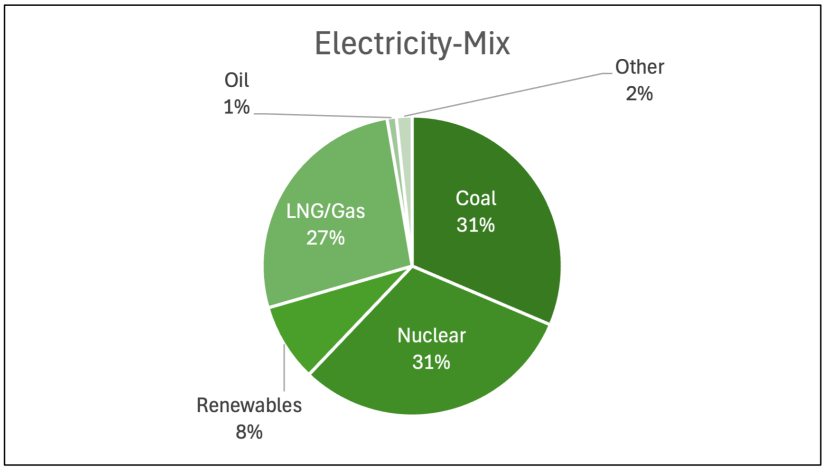

Figure 2: Electricity generation mix used for Seoul's electricity supply by energy source.

Nuclear energy contributes approximately 19% of total primary energy supply, while renewable energy sources account for only about 3% [20, 37]. Oil-derived petroleum products are almost exclusively used in the transport sector, whereas coal and nuclear energy are primarily allocated to electricity generation.

Electricity consumption totals approximately 1,769 GWh and is distributed across appliances and plug loads (45%), transport (20%), cooling (15%), lighting (12%), and shared facilities (8%) [36]. Due to upstream generation inefficiencies documented in international energy balances, electricity-related losses substantially increase overall primary energy demand [20].

The transport sector remains largely fossil-based. Internal combustion engine vehicles operate at average efficiencies of around 25%, whereas electric vehicles reach approximately 80% efficiency [17]. Despite a lower share of final energy input, electric transport provides around 42% of useful mobility, while fossil transport accounts for approximately 58% [35]. This illustrates the significant efficiency advantage of electrification.

Space heating and domestic hot water (DHW) represents the largest final energy demand at approximately 1,407 GWh. For the district 84% of heating need is covered by gas boilers and the remaining 16% of heating is covered by the district heating network. Electricity demand for cooling amounts to around 265 GWh. Assuming an air-conditioning coefficient of performance (COP) of three, this corresponds to a cooling demand of approximately 795 GWh [20].

Overall, the energy system is characterized by high fossil fuel dependency, substan-

tial conversion losses, and limited renewable penetration. The analysis indicates that electrification of transport and heating, combined with a cleaner electricity mix, offers significant potential to reduce primary energy demand while maintaining useful energy services.

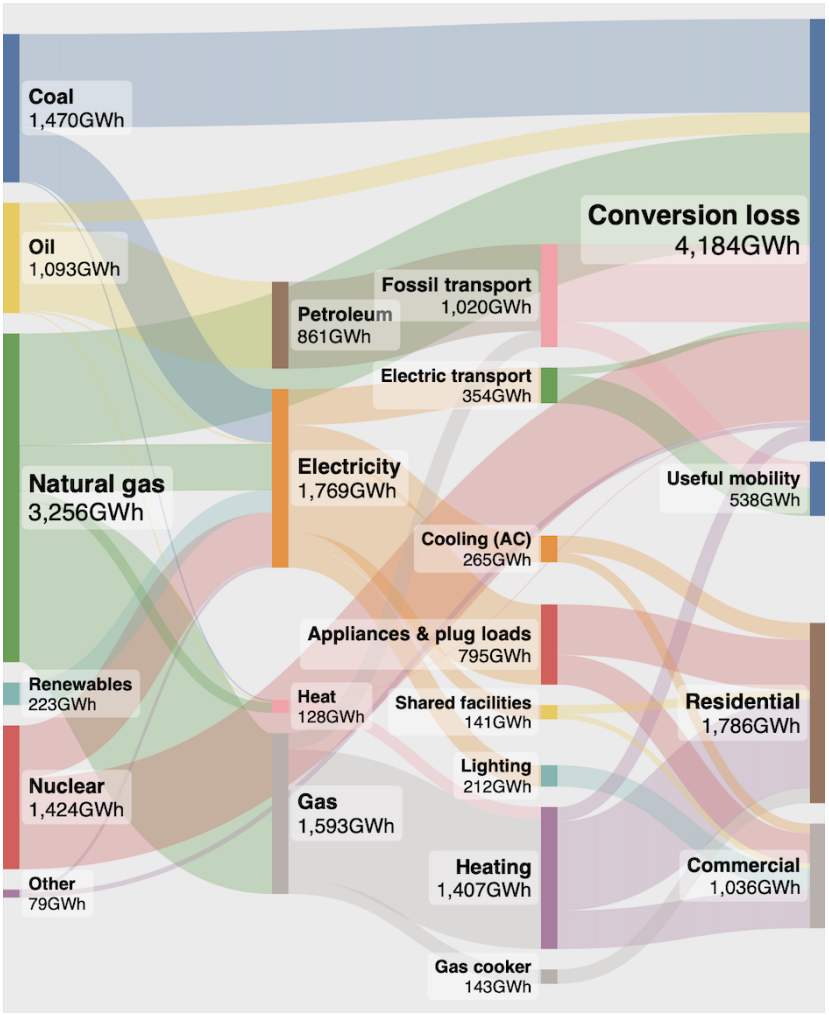

Figure 3: Primary-to-final energy flow and sectoral distribution of useful energy (2023).

1.3 Carbon emissions in Gangdong-gu

Carbon emissions in Gangdong-gu are based on the 2018 greenhouse gas inventory, which serves as the baseline for the district's Carbon Neutrality and Green Growth Plan (2025–2034). In 2020, Gangdong-gu's management-scope greenhouse gas emissions amounted to approximately 1,359 $ktCO_2eq$. The building sector dominated, accounting for about 71% of emissions (approximately 971 $ktCO_2eq$), followed by transport at around 17% (about 233 $ktCO_2eq$) and waste at roughly 11% (approximately 155 $ktCO_2eq$), while agricultural and industrial emissions were negligible. The

core structural challenge is therefore building energy use—particularly natural gas for space heating and electricity demand in high-density apartment complexes—making decarbonization primarily a building energy transition issue rather than an industrial mitigation problem [13].

1.4 Systems Mapping & System Boundary

The system boundary, illustrated in Figure 4, defines which flows and sectors are included in the analysis. The boundary is primarily geographical — encompassing all activity within Gangdong-gu — but extends beyond the physical district in one key respect: the energy system.

For the **building environment**, the boundary covers residential and commercial buildings, data centers, and their associated heating, cooling, and electricity demand. Technologies such as district heating and cooling (DHC), heat pumps, heat recovery, and retrofitting are considered within this boundary. South Korea's zero-energy building (ZEB) policies, which mandate increasing on-site energy generation and efficiency improvements, are also accounted for [2].

For **transport**, the boundary includes trips made by residents within the district and the portions of public transport routes — metro, buses, and bicycles — that operate inside Gangdong-gu. Private vehicles, EV charging infrastructure, and car-sharing are included on the same basis.

For **water management**, **waste management**, and **land use**, the boundary is strictly geographical. Water consumption, sewage treatment, and rainfall management are considered as they occur within the district. Waste generation, sorting, and treatment flows originating inside Gangdong-gu are included, even where physical treatment occurs at metropolitan-scale facilities outside the boundary. Land use — including green zones, urban cooling, and plant carbon sinks — is assessed within the district footprint.

Although the geographical focus of the study is Gangdong-gu, the electricity-system boundary extends beyond the district because electricity consumption is physically supplied through South Korea's national grid, which remains highly dependent on imported fossil fuels, with renewables contributing only around 8% of the generation mix [18]. Since the grid emission factor and electricity mix directly determine the car-

bon intensity of electrified end uses within the district, the national generation mix is treated as an explicit system input. Furthermore, renewable electricity generation, particularly solar PV, is not restricted to local rooftop installations, as external renewable capacity connected through the existing transmission network is considered in the more ambitious scenarios as a grid-linked decarbonisation measure rather than purely local on-site generation.

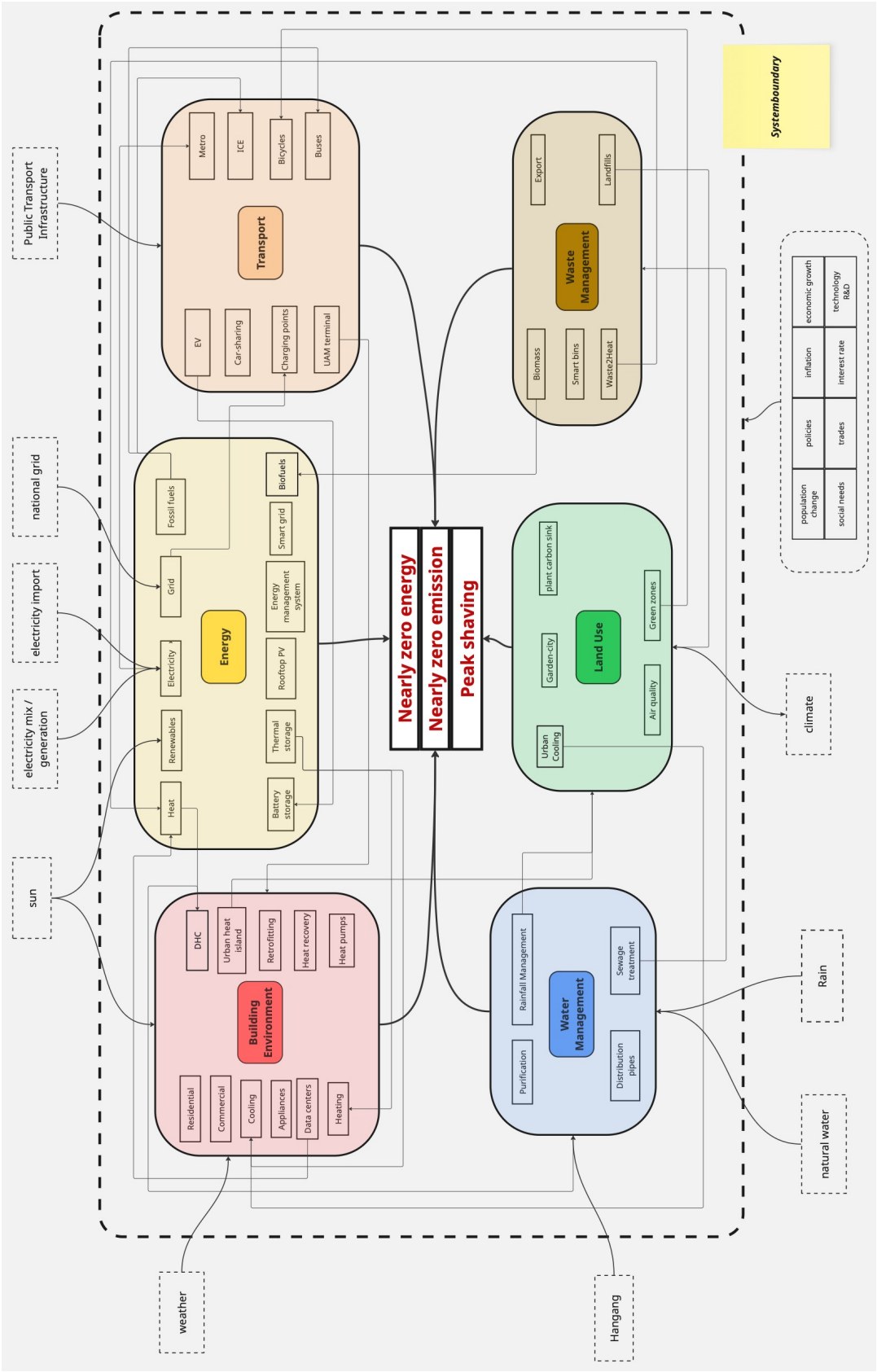

Figure 4: Integrated system mapping of Gangdong-gu.

2 Key Performance Indicators

2.1 1st KPI: "nearly" zero energy

"Nearly zero energy" is an operational (use-phase) KPI measuring the net annual delivered energy balance of the district. This balance equals total imported energy (purchased electricity, district heating, or fuels) minus creditable contributions within the system boundary, such as on-site renewable generation and recovered energy streams. The "nearly" qualifier acknowledges that absolute zero is rarely achievable in practice due to temporal supply-demand mismatches, system losses, and techno-economic constraints; the goal is instead to minimise net external energy dependence within the defined boundary conditions and available technologies.

2.2 2nd KPI: "nearly" zero emissions

"Nearly zero emissions" is defined as a district-level performance target consistent with Seoul's 2050 carbon-neutrality pathway, in which total annual GHG emissions within the system boundary approach zero. This is achieved through two coupled interventions: demand-side measures (building retrofits, appliance efficiency, demand management) that reduce final energy consumption, and supply-side decarbonisation (distributed PV, electrification, district heating decarbonisation) that lowers the carbon intensity of remaining supply. Residual emissions are limited to hard-to-abate sources such as specific commercial activities, waste-related emissions, or backup generation, and are quantified under a transparent district-level accounting framework consistent with Seoul's inventory methodology.

2.3 3rd KPI: Peak-Shaving

2.3.1 Motivation and problem framing

To complement the ongoing Zero Energy and Zero Emission targets, this project introduces **Peak Shaving** as a third critical KPI. Peak shaving aims to reduce the maximum grid import at the point of common coupling (PCC), thereby mitigating peak-time distribution network stress and potentially reducing demand-related charges.

2.3.2 KPI definition and interpretable supporting indicators

Within the PCC boundary, let the native load be $P_{\text{load}}[k]$. Let $P_{\text{ch}}[k]$ and $P_{\text{dis}}[k]$ denote battery charging and discharging power, respectively. The grid import is defined as:

$$P_{\text{grid}}[k] = P_{\text{load}}[k] - P_{\text{dis}}[k] + P_{\text{ch}}[k]. \tag{1}$$

The baseline and peak-shaved maxima over an evaluation window are:

$$P_{\text{peak},0} = \max_k P_{\text{load}}[k], \qquad P_{\text{peak,PS}} = \max_k P_{\text{grid}}[k]. \tag{2}$$

We report both absolute and relative peak shaving KPIs:

$$\Delta P = P_{\text{peak},0} - P_{\text{peak,PS}}, \qquad r = \frac{\Delta P}{P_{\text{peak},0}}. \tag{3}$$

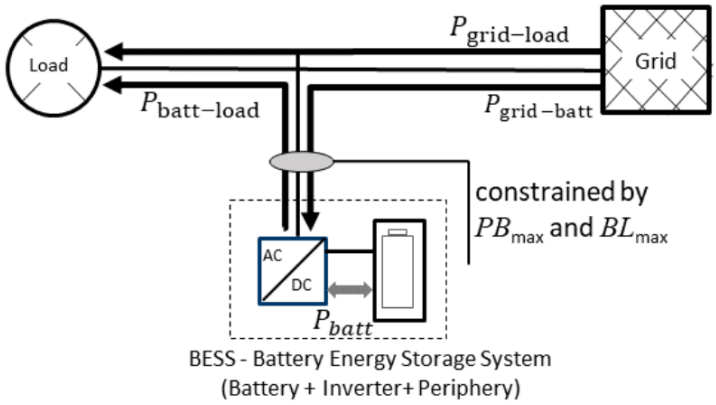

Figure 5: BESS power flow diagram for peak shaving [1].

3 Quantitative Modelling and Scenarios

3.1 Building Modelling Approach

The district-level energy modelling was performed using the optimisation software nPro. The model was constructed using 48 representative buildings to describe the district while keeping the model complexity manageable. Building usage types were determined through visual inspection using Google Maps and classified into residential, retail, supermarket, office, restaurant, hospital, and school categories. The number of representative buildings was selected as a pragmatic compromise between spatial resolution and computational complexity.

Building floor areas were estimated using two approaches depending on the building type. For standard buildings, Google Maps aerial imagery was used to estimate the building footprint. For large high-rise residential complexes, floor area data was obtained from the real estate database Zippoom. For buildings assessed through Google Maps, the following assumptions were applied to derive the net heated floor area:

- 80% of the measured plot footprint was assumed to be covered by the building structure.
- 60% of the gross floor area was assumed to constitute usable and conditioned floor area.
- The resulting floor area was scaled by the estimated average number of floors in the respective zone.

The functional distribution of building usage was estimated based on Google Maps observations and the resulting allocation across usage categories is summarised in Table 2. Residential buildings dominate the model with a share of 70%, reflecting the residential character of Gangdong-gu. Retail, supermarkets, offices, restaurants, hospitals, and schools are included to represent the mixed-use structure of the district and its non-residential electricity, cooling, and hot-water demand.

Table 2: Building zone distribution used in the nPro model

Zone	**Share [%]**
Residential	70
Retail	10
Supermarket	5
Office	6
Restaurant	5
Hospital	2
School	2

For each zone, specific annual energy demand values were assigned for space heat-

ing, domestic hot water, space cooling, and plug loads. These values were entered into the nPro general template and represent the annual demand intensity of each building function, see Table 3. The different demand intensities reflect the distinct usage patterns of the building types. For example, restaurants and supermarkets show high plug-load and cooling intensities due to kitchen equipment, refrigeration, and extended operating hours, while residential buildings are dominated by space heating and domestic hot water demand.

Table 3: Zone specific annual energy demand intensities used in the nPro template

Zone	**Space heating** [kWh/m^2/a]	**DHW** [kWh/m^2/a]	**Space cooling** [kWh/m^2/a]	**Plug load** [kWh/m^2/a]
Residential	60	28	3	34
Retail	45	5	30	90
Supermarket	35	8	60	217
Office	55	6	33	80
Restaurant	55	60	70	280
Hospital	115	45	70	200
School	20	10	25	103

For the 2035–2050 modelling period, the boundary conditions provided as part of the assignment were applied to the building demand model and scenario analysis.

This representative modelling approach allows the district to be described through a limited number of building categories while still preserving the main differences in energy demand structure. It also enables scenario analysis by applying future demand changes, technology boundaries, and energy system configurations consistently across the same underlying building demand model.

3.2 Current State Model and Validation

The current state model was developed to represent the 2025 baseline energy system of Gangdong-gu and to provide a consistent reference case for the scenario analysis. The

technology mix applied in the model follows the energy supply and demand structure described in Section 1.2. The annual energy demand breakdown used in the current state model is presented in Table 4.

For the current state model, a grid emission factor of 400 gCO_2eq/kWh was adopted in accordance with [26]. Using the adopted energy demand assumptions, the model estimates total emissions of 1323.4 $ktCO_2eq$. This corresponds to approximately 97% of the reported 2020 greenhouse gas emissions of Gangdong-gu. The deviation is considered acceptable given the simplified representation of building types, transport fuels, public transport electrification, and aggregated demand points in nPro. Therefore, the model is considered sufficiently representative for scenario comparison and strategic assessment.

It should be noted that petroleum consumption is kept consistent with the reference literature, as nPro does not directly support the modelling of conventional vehicle fuel usage. Fossil-fuel mobility is therefore represented outside the core nPro optimisation and included in the final assessment through post-processing. In addition, all public transportation, including buses and metro, is assumed to be already electrified for simplification purposes. This assumption avoids double-counting transport electricity demand, as existing electric public transport is treated as part of the current electricity demand.

Table 4: Annual energy demand in the nPro model, 2025

Energy type	**Energy [GWh]**
Petroleum	861
Space heating	1,256
Domestic hot water	538
Total thermal	**1,794**
Space cooling	389
Process cooling	403
Electricity	**1,524**

3.3 Scenarios

To evaluate possible transition pathways for the Gangdong district energy system, three scenarios with increasing levels of electrification, renewable integration, and decarbonisation ambition were developed. The scenarios are intended to represent a progression from a conservative continuation of current development trends towards a highly integrated low-carbon energy system.

Scenario 1 – Business as Usual (BAU) represents a conservative transition pathway based primarily on existing national climate targets, current technology trends, and realistic adoption rates of renewable and electrified technologies. The scenario assumes moderate reductions in fossil fuel dependence while maintaining a relatively large share of conventional technologies such as gas boilers. The purpose of this scenario is to provide a baseline reference case against which the more ambitious transition pathways can be compared.

Scenario 2 – District Energy Consideration (DEC) represents a more active district-level transition strategy focused on local energy system optimisation and electrification within the geographical district boundary. The scenario introduces 5th Generation District Heating and Cooling (5GDHC), expanded thermal storage, increased renewable integration, and higher electrification rates.

Scenario 3 – Grid-Linked Decarbonization (GLD) represents the most ambitious transition pathway, assuming strong external grid decarbonisation combined with extensive electrification and renewable deployment. The scenario removes fossil fuel-based heating entirely and allows larger renewable installations both within and outside the district boundary. Additional flexibility is introduced through expanded thermal and battery storage systems.

Together, the three scenarios provide a structured framework for analysing how different levels of technological ambition, electrification, and renewable integration influence the future energy system performance of Gangdong.

3.4 Technologies

The selected technologies were chosen because they directly address the three project KPIs: renewable generation and heat pumps reduce primary energy and emissions, 5GDHC improves system efficiency through heat recovery, and storage technologies

support peak shaving and operational flexibility.

3.4.1 PV

An installed rooftop PV capacity of 200 MW is considered realistic for Gangdong by 2050. Using typical urban assumptions, rooftop area corresponds to roughly 20–30% of total land area, of which 40–60% is suitable for PV installation. This gives an estimated usable area:

$$24.58 \times 0.25 \times 0.4 \approx 2.25 \text{ km}^2$$

Assuming an installed power density of approximately 170 W/m^2 [10], the technical potential becomes:

$$2.25 \times 10^6 \times 170 \approx 382.5 \text{ MW}$$

The assumed 200 MW in Scenario BAU therefore utilises only a reserved fraction of the technical potential and is considered a conservative but achievable deployment target (roof availability, ownership fragmentation, shading, and implementation risk). Scenario DEC raises this to 250 MW within the district boundary, while Scenario GLD adds a further 200 MW outside the district boundary, yielding a total limit of 450 MWp.

3.4.2 Geothermal

Geothermal energy has been successfully implemented at scale in Seoul [40]; however, much of Gangdong is already densely built, meaning deployment would primarily rely on retrofitting existing buildings. While such applications are technically feasible, a conservative estimate of approximately 20 MW_{th} installed geothermal capacity is assumed for Scenario BAU by 2050. Scenarios DEC and GLD raise this limit to 40 MW_{th} and 60 MW_{th}, respectively, reflecting progressively higher but still feasible levels of retrofitting across the district.

3.4.3 Battery Energy Storage System

A centralised battery energy storage system (BESS) is selected for electricity-side flexibility to reduce peak grid import. LFP chemistry is chosen over NMC and NCA due to its superior fire safety — the dominant constraint for a urban BESS in a high-density residential district. Korea has recorded multiple ESS fire incidents associated with

NMC/NCA systems, directly affecting public acceptance, insurance, and permitting risk in Seoul [30]. LFP's thermal runaway onset temperature of approximately 270 °C versus 150 °C to 200 °C for NMC, combined with longer cycle life (3,000–6,000 cycles to 80% capacity) and lower cost, make it the most suitable chemistry where safety and permitting risk outweigh energy density considerations [15, 30].

Samsung SDI's SBB 2.0 is selected as the reference system for its fire-safety positioning, domestic manufacturing base, and integrated safety stack — specifically its No Thermal Propagation (No TP) and Enhanced Direct Injection (EDI) technologies — which is the decisive differentiator for this urban site [33].

To assess realistic BESS capacities for a district-scale urban energy system, recent European utility-scale BESS projects were used as benchmark references. Goldbeck Solar started the implementation of two BESS plants in Germany with a combined power rating of approximately 100 MW and a combined storage capacity of approximately 300 MWh. The projects are designed for a three-hour operating cycle and are connected directly to the high-voltage grid through 110 kV substations [22]. This benchmark shows that storage capacities in the range of several hundred MWh are already being implemented as grid-connected infrastructure projects.

A second reference is Harmony Energy's Foriaux BESS project in France, which has a rated capacity of 100 MW/200 MWh and represents the company's third project of this size brought to construction stage in France [29]. The project corresponds to a two-hour storage duration.

These benchmark projects support the plausibility of the maximum storage capacities of 200 MWh and 300 MWh selected in this study for the nPro optimization process. Therefore, both selected maximum BESS configurations are within the range of currently implemented European grid-scale storage projects.

3.4.4 Fifth-Generation District Heating and Air-Source Heat Pumps

In 2025, the South Korean government officially recognized air-source heat as a renewable energy source and introduced a dedicated electricity tariff plan for heat pump systems [9]. This policy shift reflects the increasing importance of electrified heating technologies in reducing fossil fuel dependence and greenhouse gas emissions.

Air-source heat pumps offer significantly higher energy efficiency compared to direct

electric heating systems because they transfer thermal energy from ambient air rather than generating heat solely from electricity. Typical coefficients of performance (COP) for ASHPs are commonly between 2 and 4, meaning that one unit of electricity can provide multiple units of thermal energy. This makes ASHPs an attractive solution for both residential and commercial applications in Korea's decarbonization strategy.

However, in dense urban districts, decentralized ASHP deployment may face challenges related to electrical peak demand, outdoor unit placement, urban noise, and limited installation space. In these contexts, fifth-generation district heating and cooling (5GDHC) systems present an alternative approach with potentially higher system-level efficiency [5, 27].

Unlike conventional district heating systems, 5GDHC networks operate at near-ambient temperatures and utilize decentralized heat pumps within buildings. These systems allow bidirectional heat exchange between connected buildings, enabling simultaneous heating and cooling demands to be balanced across the district. Waste heat from commercial buildings, data centers, metro systems, or cooling-dominated buildings can therefore be recovered and reused for space heating and domestic hot water production [19].

The integration of both heating and cooling loads is particularly advantageous in dense mixed-use urban environments, where residential buildings often require heating while office and commercial buildings require cooling depending on the season. Under such conditions, 5GDHC systems can achieve higher overall energy efficiency than standalone ASHP systems because internal heat recovery reduces external energy demand and decreases peak electricity consumption [6].

Furthermore, fifth-generation district systems can integrate additional low-grade renewable and waste heat sources such as geothermal energy, seawater heat exchange, and seasonal thermal energy storage. This flexibility improves long-term resilience and supports large-scale electrification while mitigating stress on the electricity grid [21].

For South Korea, where urban density is among the highest in the world, the combination of widespread ASHP adoption and future implementation of low-temperature district heating networks could provide a complementary pathway toward sustainable urban heating and cooling.

3.4.5 Biofuels and Biomass

According to [13], the local production capacity from the Gangdong-gu resource recycle center targets 3.16 million Nm^3/year of biogas from 2034. Based on a lower heating value (LHV) for biogas of 6.5 kWh/Nm^3 [4], the theoretical maximum energy producible locally within the district amounts to approximately 20.54 GWh. Biomass feedstocks in South Korea span agricultural and forestry residues as well as imported wood pellets, though policy has increasingly favoured domestic woody biomass [16]. Wood pellets — both imported and domestic — remain the dominant feedstock: South Korea ranks as Asia's second-largest consumer of forest biomass for energy, with 3.9 million metric tons of wood pellets imported as of April 2024, of which 71% originated from Southeast Asia. Municipal solid waste (MSW) and waste-to-energy schemes are also noteworthy; the plant at Yangchun, Seoul, operated by Hankook Seghers Co. Ltd., for instance, comprises two units of 200 tonnes/day capacity commissioned in 1996 [32]. Coal co-firing with biomass represents a further pathway, enabling gradual integration of biomass into the energy mix without requiring entirely new infrastructure. On 18 December 2024, the South Korean government announced a substantial reform of public support for biomass energy, phasing down indirect subsidies in the form of Renewable Energy Certificates (RECs) for most biomass categories. From 2025 onwards, new biomass power plants will no longer qualify for RECs, though plants already under construction or in planning with approved business permits are exempt and subject to phased reduction timelines. This has been widely characterised as the largest biomass policy rollback of its kind in Asia. Domestic biomass is partially exempted: REC weightings for pellets and chips produced from domestic forestry residues remain unchanged at 1.5 for co-firing and 2.0 for dedicated combustion. In light of these policy shifts and the limited domestic production capacity relative to district-level demand, biofuels are not considered a viable option and are excluded from the modelling.

3.5 Demand Assumptions

3.5.1 Building Energy

The demand changes from the current state to 2050 were considered identical across all scenarios. The assumptions were derived from articles on estimated demand changes

[23], efficiency improvements of new buildings and retrofitting, sales trends in air conditioning units [31], population projections, and anticipated climate-change-driven temperature increases. The adopted values are presented in Table 5.

Table 5: Energy demand assumptions for different end uses in 2050

End use	**2025**	**2050**
Space heating	100%	90%
Domestic hot water	100%	90%
Space cooling	100%	125%
Process cooling	100%	100%
Plug loads	100%	110%

3.5.2 E-Mobility

As a simplifying assumption, all vehicles that are already electric in the current system are treated as public vehicles and are assumed to be already included in the existing plug-load electricity demand. Therefore, only the additional private EV uptake in the 2050 scenarios is added as new e-mobility electricity demand in nPro. The annual private EV electricity demand is calculated as:

$$E_{\mathrm{EV}} = N_{\mathrm{cars},2025} \cdot (1 - r_{\mathrm{car}}) \cdot s_{\mathrm{EV}} \cdot d_{\mathrm{annual}} \cdot e_{\mathrm{EV}}$$

where $N_{\mathrm{cars},2025}$ is the current private car stock, r_{car} is the scenario-specific private car reduction, s_{EV} is the EV share, d_{annual} is the annual driving distance, and e_{EV} is the electricity demand per kilometre.

The e-mobility demand in nPro was derived from the projected private car stock in Gangdong-gu, the scenario-specific private car reduction, and the assumed share of electric vehicles. The current private vehicle stock in Gangdong-gu is based on Seoul's vehicle registration statistics. According to Seoul Metropolitan Government traffic data, Gangdong-gu had 168,985 registered vehicles [39]. This value is used as the 2025 baseline vehicle stock for the e-mobility demand calculation. For each scenario, this vehicle stock is reduced according to the assumed decrease in private car use by 2050.

The remaining private vehicle fleet is then split into electric and fossil-fuel vehicles based on the scenario-specific EV share.

For the BAU scenario, a 20% reduction in private cars is assumed. This results in a projected 2050 private car stock of 135,188 vehicles:

$$168{,}985 \cdot (1 - 0.20) = 135{,}188$$

The annual driving distance is assumed to be 9,000 km per vehicle. For electric vehicles, an electricity consumption of 0.18 kWh/km is assumed. Fossil-fuel vehicles are treated separately using a reference consumption of 6.5 l/100 km and an energy content of 8.8 kWh/l, corresponding to 0.572 kWh/km of final fuel energy.

For BAU, 70% of the remaining private car fleet is assumed to be electric. Therefore, the corrected annual private EV electricity demand is calculated as:

$$135{,}188 \cdot 0.70 \cdot 9{,}000 \cdot 0.18 = 153.3 \text{ GWh/year}$$

The remaining 30% of private cars are assumed to remain fossil-fuelled and are treated separately in the post-processing of fossil mobility energy demand. Using the final fuel energy intensity of 0.572 kWh/km, the corresponding fossil-fuel mobility demand is:

$$135{,}188 \cdot 0.30 \cdot 9{,}000 \cdot 0.572 = 208.8 \text{ GWh/year}$$

This annual e-mobility demand is then uploaded to nPro as an additional electricity load. The hourly distribution of this load follows the scenario-specific EV charging boundaries described in Section 3.5.3. In this way, the model captures not only the annual increase in electricity demand from transport electrification, but also the effect of different charging strategies on peak electricity demand.

3.5.3 EV Charging Boundaries

The EV load profiles uploaded to nPro differ across the scenarios in Table 6. BAU assumes no specific charging boundaries and therefore represents largely uncontrolled charging behaviour without defined charging windows. DEC introduces coordinated charging, with 20% public charging between 08:00 and 17:00 and 80% private residential charging between 19:00 and 06:00. GLD further shifts charging towards residential overnight charging, with 10% public charging between 08:00 and 17:00 and 90% private charging between 23:00 and 07:00.

For DEC and GLD, the charging loads are not modelled as fixed deterministic loads within these time windows. Instead, a natural stochastic probability distribution was created to represent realistic charging behaviour. This means that charging events are distributed probabilistically within the defined charging windows rather than occurring simultaneously at the start of each window. The distribution accounts for typical user behaviour, such as different arrival times, charging start times, and charging durations. Workday and weekend patterns are also considered, since public and residential charging behaviour differs depending on daily mobility routines.

This approach avoids unrealistic load spikes and provides a more representative EV demand profile for the district. As a result, the model can better assess the impact of coordinated charging on peak electricity demand and the effectiveness of peak-shaving measures in DEC and GLD.

Table 6: EV charging boundary assumptions used in the nPro load profiles

Charging parameter	**BAU**	**DEC**	**GLD**
Charging boundary applied	No	Yes	Yes
Public charging share	–	20%	10%
Public charging window	–	08:00–17:00	08:00–17:00
Private charging share	–	80%	90%
Private charging window	–	19:00–06:00	23:00–07:00
Workday/weekend differentiation	No	Yes	Yes
Stochastic charging behaviour	Uncontrolled profile	Natural profile	Natural profile

The effect of the charging boundary assumptions is illustrated in Figure 6. The figure compares the resulting EV charging load profiles across all three scenarios. In BAU, the absence of defined charging windows leads to a less coordinated charging pattern. In DEC, the separation between daytime public charging and evening or overnight residential charging distributes the EV load more systematically across the day. In GLD, the stronger shift towards private overnight charging moves a larger share of charging demand into late-night hours, reducing the overlap with evening household electricity peaks. This comparison shows how increasingly coordinated charging boundaries can

reduce peak-load pressure despite higher EV shares without any BESS considered for now.

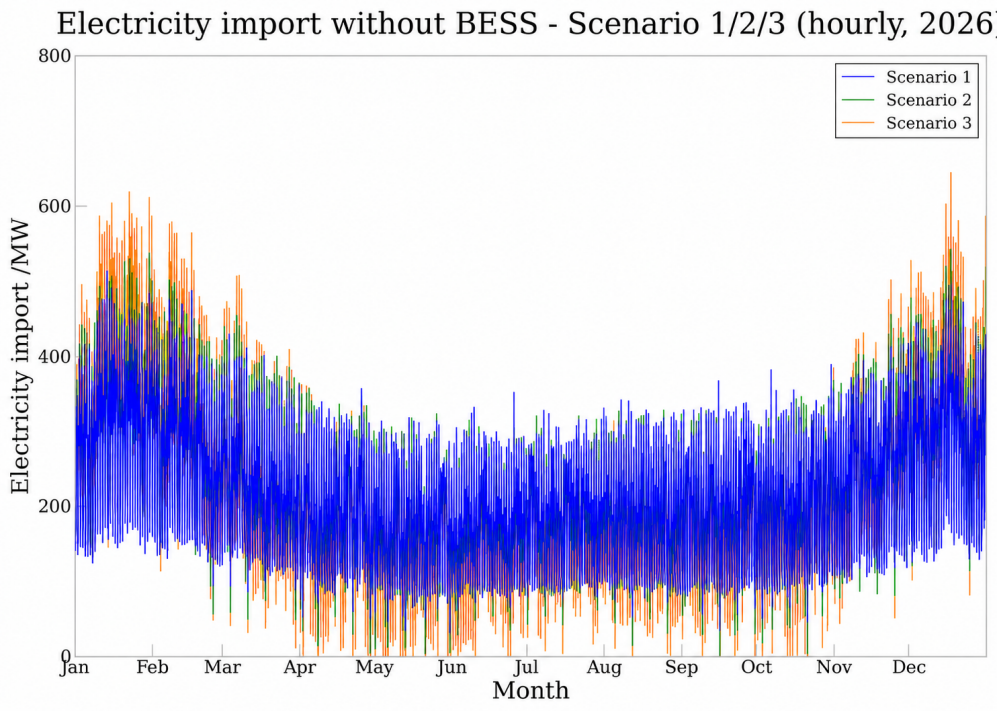

Figure 6: Comparison of electrical grid import under the scenario-specific charging boundary assumptions.

3.6 Economic and Technical Input Parameters

The techno-economic input parameters used in nPro define investment costs, lifetimes, maintenance costs, energy carrier prices, emission factors, and primary energy factors, applied consistently across all scenarios as the basis for life-cycle cost calculation, technology selection, and environmental performance assessment.

Network temperature assumptions differ by system generation. The 4GDH network operates at fixed supply and return temperatures of 55 °C and 25 °C respectively. The 5GDHC network uses an ambient-temperature-dependent curve, with supply temperature decreasing from approximately 45 °C at low ambient temperatures to 15 °C at high ambient temperatures, and return temperature from 25 °C to 7 °C. This near-ambient operation improves integration of heat pumps, geothermal sources, waste heat, and cooling demand, as summarised in Table 7.

Table 7: District heating and cooling network temperature assumptions implemented in nPro

Network type	Ambient air temp.	Supply temp.	Return temp.
4GDH	–	55 °C	25 °C
5GDHC	-10 °C	45 °C	25 °C
5GDHC	25 °C	15 °C	7 °C

Technology cost, lifetime, and maintenance assumptions are shown in Table 8. Higher CAPEX or shorter lifetimes increase the effect on total life-cycle costs and stakeholder economics.

Table 8: Technology cost, lifetime, and maintenance assumptions used in nPro

Technology	Specific CAPEX [USD/kW or USD/m^3]	Lifetime [years]	Maintenance cost [% of CAPEX/a]
PV	650	25	1.5
Battery	180	15	2.0
Heat storage [m^3]	90	30	1.0
Wind onshore	1350	30	3.0
ASHP	550	20	2.5
Geothermal shallow	800	50	1.0
Cold storage [m^3]	120	25	1.0
Geothermal heat pump	700	20	2.5
Heat recovery chiller	700	20	3.0
Geothermal chiller	700	20	2.5

Imported energy carrier assumptions are listed in Table 9. Electricity import carries the highest non-renewable primary energy factor and a CO_2 emission factor of 210 gCO_2/kWh, while district heating, cooling, and biogas have lower emission and

primary energy factors.

Table 9: Supply price, emission factor, and primary energy factor assumptions used in nPro

Energy carrier	**Supply price [USD/kWh]**	**CO_2 emissions [gCO_2/kWh]**	**PEF non-renewable [-]**
Electricity import	0.15	210	1.7
District heat import	0.10	160	0.7
District cooling import	0.10	45	0.4
Biogas	0.10	35	0.2

Global boundary conditions used for modelling and post-processing are summarised in Table 10, based on the project boundary condition data provided in the lecture material.

Table 10: Boundary conditions used in the modelling and post-processing

Parameter	**Value**
Renewable electricity fraction	35%
Grid emission factor	210 gCO_2eq/kWh
Electricity buy price	0.15 USD/kWh
Electricity sell price in nPro	0 USD/kWh
Project lifetime	15 years
Start year of study period	2035
Discount rate	5%
Inflation rate	2%

3.7 Maximum Technology Capacities and Scenario Screening

It should be noted that nPro primarily optimises the energy system from a techno-economic perspective. The three project KPIs were therefore not all direct optimisa-

tion objectives. While emissions and primary energy demand can be influenced through input parameters and technology constraints, peak shaving had to be evaluated externally using the hourly electricity import profiles. The KPI results should therefore be understood as combined outcomes of nPro optimisation, scenario constraints, and post-processing calculations.

Because nPro is an optimisation tool rather than a pure simulation platform, technology deployment cannot be fully prescribed for DEC and GLD. Instead, upper capacity limits were defined for each technology, and the optimiser was allowed to select any installed capacity up to these limits. This approach makes it possible to compare different technology pathways while still allowing nPro to identify cost-effective system configurations.

Since nPro does not directly optimise peak shaving as an independent objective, an electricity capacity import price was used as a modelling proxy to incentivise battery storage and flexible operation. The resulting BESS capacities and hourly electricity-import profiles were then evaluated in post-processing to quantify the peak-shaving KPI.

To explore technology combinations that may not be selected under pure cost optimisation, selected technologies were additionally forced into the model in dedicated runs. This allowed otherwise sub-optimal but strategically relevant configurations to be assessed and compared in terms of cost, emissions, renewable integration, and peak-shaving performance. The maximum capacity limits used across all three scenarios are summarised in Table 11.

Table 11: Maximum technology capacities used as upper bounds in the scenario optimisation

Parameter	**BAU (max.)**	**DEC (max.)**	**GLD (max.)**
DH	25% (4th Gen)	30% (5GDHC)	50% (5GDHC)
ASHP decentralised	25%	60%	50%
PV [MWp]	200	250	450
Wind onshore [MW]	–	–	50
Geothermal [MW_{th}]	20	40	60
BESS centralised [MWh]	–	200	300
Cooling tank [m^3]	–	–	10,000
Heat tank [m^3]	–	10,000	20,000
Accumulator tank [m^3]	–	10,000	20,000
Biogas [GWh]	–	100	200
CHP [MW_{el}]	–	100	150
ASHP centralised [MW_{th}]	–	100	150
EV share	70%	80%	90%
Private car reduction	20%	30%	40%
Charging boundaries	No	Applied	Applied

For BAU, the technology values were treated as fixed deployment assumptions rather than optimisation limits, as the scenario is intended to reflect a realistic continuation of current trends. For DEC and GLD, multiple optimisation runs were performed under different objective functions, including cost minimisation, CO_2 reduction, primary-energy reduction, and peak-shaving improvement. The resulting solutions were used to construct Pareto fronts for the main trade-offs between economic performance and the three project KPIs.

The portfolio screening was carried out in three steps. First, the trade-off between the three technical KPIs was analysed. Reducing CO_2 emissions and primary energy

demand generally requires higher electrification, larger renewable energy capacities, and stronger district-energy integration. However, these measures can also increase peak electricity demand. Therefore, peak shaving becomes a critical third KPI, ensuring that decarbonisation and electrification do not simply shift the burden from fossil fuel use to electricity-grid stress.

Second, the trade-off between technology ambition and economic performance was evaluated. More ambitious technology portfolios can improve CO_2 emissions, primary energy demand, and peak-shaving performance, but they usually require higher CAPEX, additional storage, more complex operation, and therefore lower economic attractiveness. This trade-off is captured through Pareto-front diagrams comparing NPV with CO_2 emissions, NPV with primary energy demand, and NPV with peak-shaving performance.

Third, an implementation-risk filter was applied. The selected solutions should not only perform well in the model, but also remain realistic under district-level constraints. These constraints include limited available area for technology installation, regulatory uncertainty, permitting complexity, grid connection limitations, and operational complexity. As a result, the final portfolio selection prioritises solutions that provide strong KPI improvements while avoiding technologies with high implementation risk.

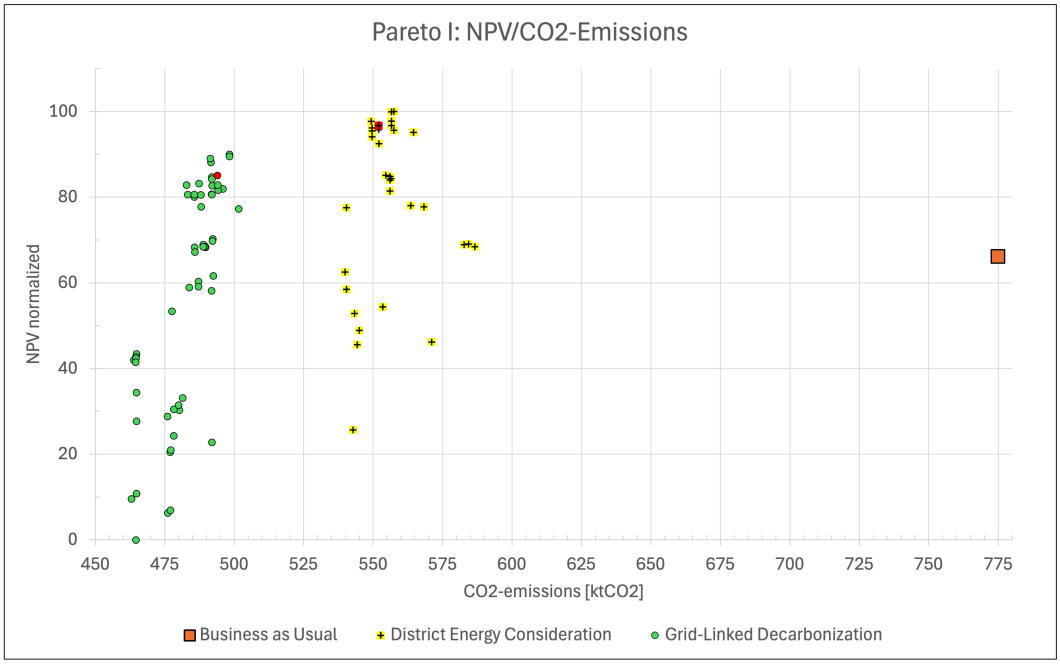

Figure 7: Pareto front showing the trade-off between net present value (NPV) and CO_2 emissions for DEC and GLD.

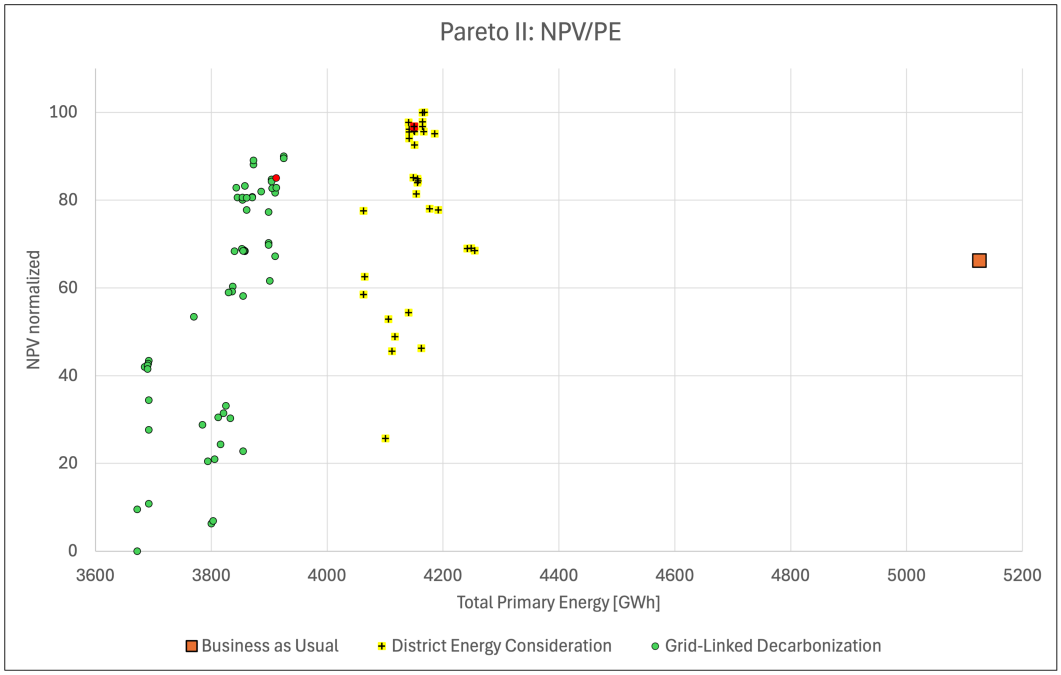

Figure 8: Pareto front showing the trade-off between net present value (NPV) and primary energy demand for DEC and GLD.

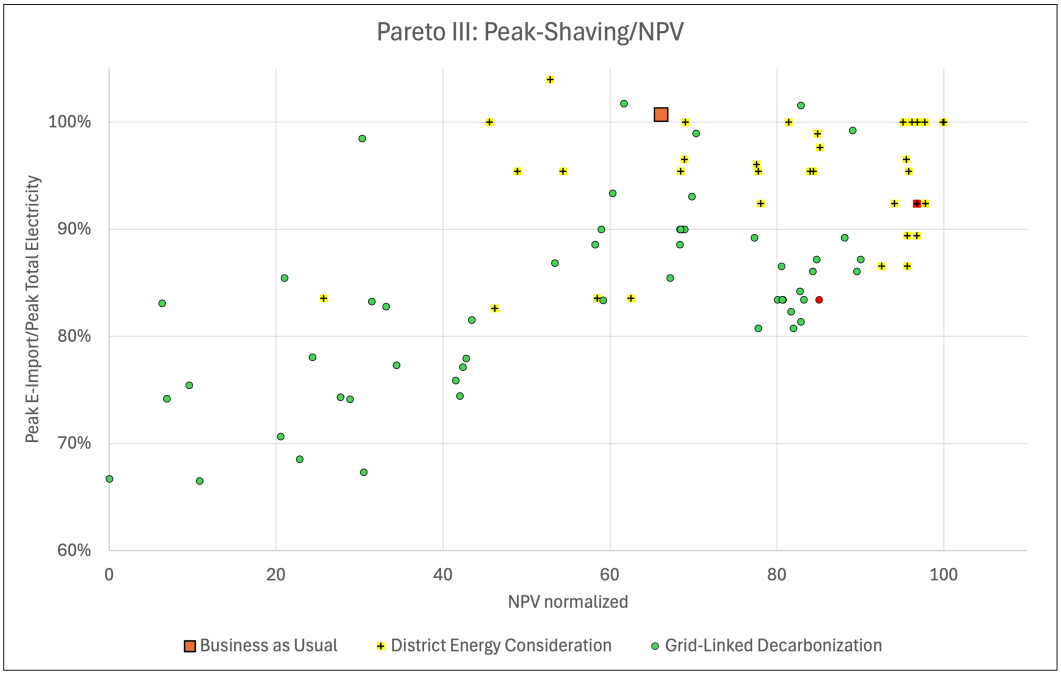

Figure 9: Pareto front showing the trade-off between net present value (NPV) and peak-shaving performance for DEC and GLD.

The selected portfolio cases correspond to the red dots in Figures 7–9. These cases were selected because they provide a balanced improvement across all three KPIs while remaining economically and practically feasible. The objective was not to identify the

single lowest-emission or highest-NPV solution, but to find a robust compromise between CO_2 reduction, primary-energy reduction, peak-shaving performance, technology cost, and implementation risk.

Several technologies were excluded from the final selection despite appearing in some optimisation runs. Biogas was excluded because the district does not possess sufficient local production capacity to supply the modelled demand at scale, and national support for biomass-based generation is being reduced. Wind power was also excluded because, compared with additional PV capacity, it faces higher installation complexity, more difficult permitting, stronger area and siting constraints, and higher implementation risk in the dense Seoul metropolitan context. Therefore, the final selected portfolios focus on technologies that are technically effective, economically reasonable, and more realistic to implement within the district boundaries.

The final selected technology configurations are presented in Table 12. They represent balanced transition pathways that improve emissions, primary energy demand, and peak-shaving performance while limiting unnecessary system complexity and implementation risk.

The maximum technology capacities in Table 11 define the optimisation boundaries, whereas Table 12 presents the final selected technology portfolios used for the detailed scenario analysis. These portfolios correspond to the selected red-dot cases in the Pareto-front figures and therefore represent the final scenario configurations evaluated in the results section. The selected portfolios are not necessarily the maximum technically possible configurations, but balanced solutions that combine KPI performance, economic feasibility, and implementation realism.

Table 12: Final selected technology portfolios used in the detailed scenario analysis. The cases correspond to the selected red dots in Figures 7–9.

Parameter	**BAU**	**DEC**	**GLD**
DH	25% (4th Gen)	30% (5GDHC)	50% (5GDHC)
ASHP	25%	60%	50%
PV rooftop [MWp]	200	250	250
PV outside district [MWp]	–	–	200
PV total [MWp]	**200**	**250**	**450**
Geothermal [MW_{th}]	20	30	40
BESS centralised [MWh]	–	100	200
Accumulator tank [m^3]	–	6,000	10,000
EV share	70%	80%	90%
Private car reduction	20%	30%	40%
Charging boundaries	No	Applied	Applied

3.7.1 Business as Usual - Final Selected Portfolio

The BAU scenario is based on the South Korean nationally determined contribution (NDC) [43] where it does not conflict with the given boundary conditions [26].

The district is projected to experience a shift in mobility patterns driven by increased environmental awareness, expanded public transport infrastructure, and a declining population. Based on these factors, a 20% reduction in private car usage is assumed for the modelling period. The NDC targets 70% of all new vehicles sold in 2035 to be electric, and accordingly 70% of all private vehicles within the district are assumed to be electric by 2050.

The heating supply mix consists of 4th generation district heating (4th Gen. DH) and ASHP, each accounting for 25%, with gas boilers representing the remaining 50%. The continued reliance on gas boilers reflects a conservative estimate consistent with Korea's near-term energy transition trajectory, and aligns broadly with the emissions

reduction targets in the NDC without assuming radical structural change at the district level. Rooftop PV is limited to 200 MWp, geothermal capacity to 20 MW_{th}, and no central BESS or accumulator tank is included. No sufficiently granular technology deployment data was available to justify a more specific allocation. Large changes in district cooling supply are not expected, as air conditioning is already widely used in South Korea [31].

3.7.2 District Energy Consideration - Final Selected Portfolio

The DEC scenario represents the preferred district-level transition pathway. It builds on the BAU assumptions by introducing stronger electrification, higher renewable integration, coordinated EV charging, and 5th generation district heating and cooling, while keeping renewable energy deployment within the district boundary. Mobility patterns shift further relative to BAU, with 30% fewer private car trips and 80% EV penetration by 2050.

The final selected DEC portfolio corresponds to the balanced red-dot case identified in the Pareto analysis. It includes 30% 5GDHC, 60% ASHP, and 10% remaining gas boilers for heating supply. Cooling demand is partly supplied by 5GDHC, while the remaining cooling demand is covered by conventional air chillers. The portfolio also includes 250 MWp rooftop PV, 30 MW_{th} geothermal capacity, a 100 MWh central BESS, and a 6,000 m^3 accumulator tank.

This configuration was selected because it provides substantial improvements across all three KPIs while remaining more realistic to implement than the most ambitious GLD pathway. It reduces fossil fuel dependence, improves system efficiency through district energy integration, and supports peak-load management through storage and coordinated charging, without relying on renewable capacity outside the district boundary.

3.7.3 Grid-Linked Decarbonization - Final Selected Portfolio

The GLD scenario represents the most ambitious decarbonisation pathway and is treated as an alternative high-ambition scenario rather than a direct extension of DEC. It assumes stronger electrification, complete removal of gas boilers, larger storage capacity, and additional renewable generation outside the district boundary. Private car

usage is reduced by 40%, and EV penetration reaches 90% by 2050, with the residual 10% assumed to remain fossil-fuelled.

The final selected GLD portfolio corresponds to the high-ambition red-dot case identified in the Pareto analysis. It includes 50% 5GDHC and 50% ASHP for heating supply, resulting in a fully electrified local heating system. Cooling demand is supplied by 50% 5GDHC and 50% conventional air chillers. The portfolio includes 250 MWp rooftop PV within Gangdong-gu and an additional 200 MWp PV outside the district boundary, resulting in 450 MWp total PV capacity. It also includes 40 MW_{th} geothermal capacity, a 200 MWh central BESS, and a 10,000 m^3 accumulator tank.

The external renewable capacity in GLD is treated as a grid-linked decarbonisation measure within the extended electricity system boundary described in Section 1.4. This allows GLD to achieve the strongest environmental performance, but also increases implementation complexity, investment requirements, and reliance on coordination beyond the district boundary.

4 Results

The results are structured based on the three scenarios presented above; Business as usual, district energy consideration and grid-linked decarbonization and is presented in terms of the three key performance indicators: Nearly zero energy, nearly zero emissions and peak shaving. Supporting economic, renewable energy, and qualitative indicators are presented alongside.

The *2025* case represents the current state of the Gangdong-gu energy system. It is not a future scenario, but the baseline against which all scenario improvements are measured. The current state reflects today's energy demand, technology mix, fossil-fuel dependency, grid reliance, and emission level. Therefore, CO_2 reduction and primary energy reduction indicators are calculated relative to this 2025 reference case.

The *2050 NoChanges* (2050NC) case represents a counterfactual asset-replacement baseline. Future demand development until 2050 is considered, but no major additional decarbonisation or system transformation measures are implemented. However, the case is not a zero-investment baseline: since most existing technologies do not last until 2050, replacement investments are included at the end of their technical lifetime.

The existing district heating network is the main exception and is assumed to remain in place. Therefore, 2050NC represents the cost of maintaining the current system logic until 2050, including necessary reinvestments.

4.1 Post-processing approach

Several parts of the assessment were calculated outside nPro in a post-processing step, because they are either not directly represented in the optimisation model or require additional scenario-specific accounting. The nPro model provides the optimised energy system configuration, annual energy balances, technology capacities, electricity imports, costs, and operational results for the building and district energy system. These outputs were then combined with external assumptions to calculate the final KPIs for primary energy demand, CO_2 emissions, peak shaving, and economic performance.

A key post-processing element concerns transport energy demand. nPro only represents the additional electricity demand from electric vehicles as part of the electricity load profile. Conventional fossil-fuel mobility is not directly modelled in nPro. Therefore, the remaining fossil-fuel vehicle demand in each scenario was calculated externally based on the projected private vehicle stock, the assumed private car reduction, the scenario-specific EV share, annual driving distance, and fuel energy intensity. The resulting fossil-fuel mobility energy demand and associated emissions were added to the final system balance.

In addition to transport energy demand, vehicle-related investment and operating costs were also included in the post-processing step. For the 2035 assessment, capital cost parity between electric vehicles and internal combustion engine vehicles was assumed, with both vehicle types assigned a CAPEX of 30,000 USD per vehicle. Both EVs and ICE vehicles were modelled with a technical lifetime of 15 years and a salvage value of 0%. Annual vehicle maintenance costs were assumed to amount to 2.5% of CAPEX for EVs and 4.0% of CAPEX for ICE vehicles. These assumptions were applied consistently across scenarios to calculate vehicle-related lifecycle costs.

Charging infrastructure costs were also added externally, since nPro only accounts for the electricity demand of EV charging and does not explicitly model private and public charging infrastructure deployment. CAPEX was estimated from the corrected

annual EV charging demand using specific costs of 0.183 million USD/GWh for private charging and 0.520 million USD/GWh for public charging. For the BAU case, the EV charging demand is 153.3 GWh, resulting in 28.1 million USD for private charging and 79.7 million USD for public charging, or 107.8 million USD in total. A lifetime of 12 years, zero salvage value, and annual maintenance costs of 4.0% of charging infrastructure CAPEX were assumed.

In addition, the renewable electricity fraction of the public grid was treated in post-processing. The project boundary conditions specify a renewable electricity fraction of 35% for the study period. Since this value is not a direct input in nPro, it was applied after the optimisation to distinguish renewable and non-renewable shares of imported grid electricity.

Peak shaving was also evaluated in post-processing. nPro was used to generate hourly electricity import profiles, while the peak-shaving KPI was calculated externally by comparing the maximum electricity import before and after the application of storage and charging strategies. This allowed the absolute peak reduction and the relative peak-shaving effect to be quantified consistently across scenarios.

Economic results were also adjusted in post-processing where necessary. In particular, DHC network costs were estimated externally, because the nPro model represents the district through fewer than 50 aggregated demand points, whereas the real building stock of Gangdong-gu contains 20,192 main buildings. As a result, the network generated by nPro does not fully capture the actual number of building connections and distribution pipes required in the district. The externally estimated DHC network costs were therefore added to the economic assessment to avoid underestimating infrastructure investment requirements.

4.2 Project KPIs

Table 13 summarises the quantitative energy and emissions outcomes for the current state of 2025, the 2050 reference case, and the three transition scenarios. All transition scenarios achieve meaningful reductions relative to the 2025 current-state baseline. GLD performs best across the energy and emissions indicators, reaching a 62% reduction in CO_2 emissions, corresponding to 494 ktCO_2, and a 45% reduction in primary energy consumption, corresponding to 3,912 GWh. The remaining emissions in

this scenario are mainly associated with residual fossil-fuel vehicle use, waste-related emissions, and the remaining electricity demand supplied by the public grid. They are therefore not primarily caused by the direct operation of the local energy system. DEC achieves intermediate results, with a 57% emissions reduction and a 42% reduction in primary energy demand. BAU delivers the smallest improvements among the transition scenarios, reducing CO_2 emissions by 40% and primary energy demand by 28%.

The peak electricity grid-import indicator requires additional interpretation. Due to increasing electrification of heating, cooling, and transport, the absolute peak electricity demand increases compared to the current state. Therefore, the peak grid-import values alone do not fully describe the effectiveness of flexibility measures such as battery storage, load shifting, and controlled EV charging. To compare the peak-shaving effect across scenarios with different electrification levels, a normalised indicator is used: *Peak-Shaving vs Peak E-Demand*. This ratio expresses the achieved peak reduction relative to the total peak electricity demand of each scenario. As a result, the indicator shows how effectively each scenario reduces its own peak load despite higher electrification. The BESS is sized from a peak-shaving scan of the hourly district electricity-import profile. Scenario DEC adopts a 50 MW/100 MWh BESS and Scenario GLD a 100 MW/200 MWh BESS; beyond these power ratings, additional capacity yields diminishing peak-shaving returns as energy capacity becomes the binding constraint. The required BESS discharge power at each hour is:

$$P_{\mathrm{BESS},t} = \min(P_{\max},\, \max(0,\, P_t - T))\,, \tag{4}$$

where P_t is the original electricity import, T is the shaving threshold, and $P_{\max}$ is the rated discharge power. Post-BESS import is $P_{\mathrm{after},t} = P_t - P_{\mathrm{BESS},t}$, and the final threshold resolves as:

$$T_{\mathrm{final}} = \max(T_{\mathrm{power}},\, T_{\mathrm{energy}}), \tag{5}$$

where $T_{\mathrm{power}} = P_{\mathrm{peak}} - P_{\max}$ and T_{energy} is the lowest feasible threshold within the rated energy capacity. This gives final thresholds of approximately 491.5 MW for Scenario DEC and 538.5 MW for Scenario GLD. BAU achieves only 2% peak shaving, while DEC reaches 14% (Figure 11) and GLD reaches 23% (Figure 12), reflecting the stronger use of storage and flexibility options in the more advanced scenarios.

Table 13: Scenario comparison of emissions, energy and grid indicators. Peak grid import refers to the final peak after smart charging and BESS operation. Peak shaving is measured relative to a reference case without smart charging and without BESS.

Parameter	**2025**	**2050NC**	**BAU**	**DEC**	**GLD**
CO_2 emissions [ktCO_2]	1,294	1,005	775	552	494
CO_2 reduction vs. 2025	–	−22%	−40%	−57%	−62%
Primary energy [GWh]	7,122	5,700	5,126	4,150	3,912
Primary energy reduction vs. 2025	–	−20%	−28%	−42%	−45%
Peak grid import after smart charging and BESS [MWp]	322	357	514	491	538
Peak shaving ratio vs. reference without smart charging and BESS	–	–	2%	14%	23%

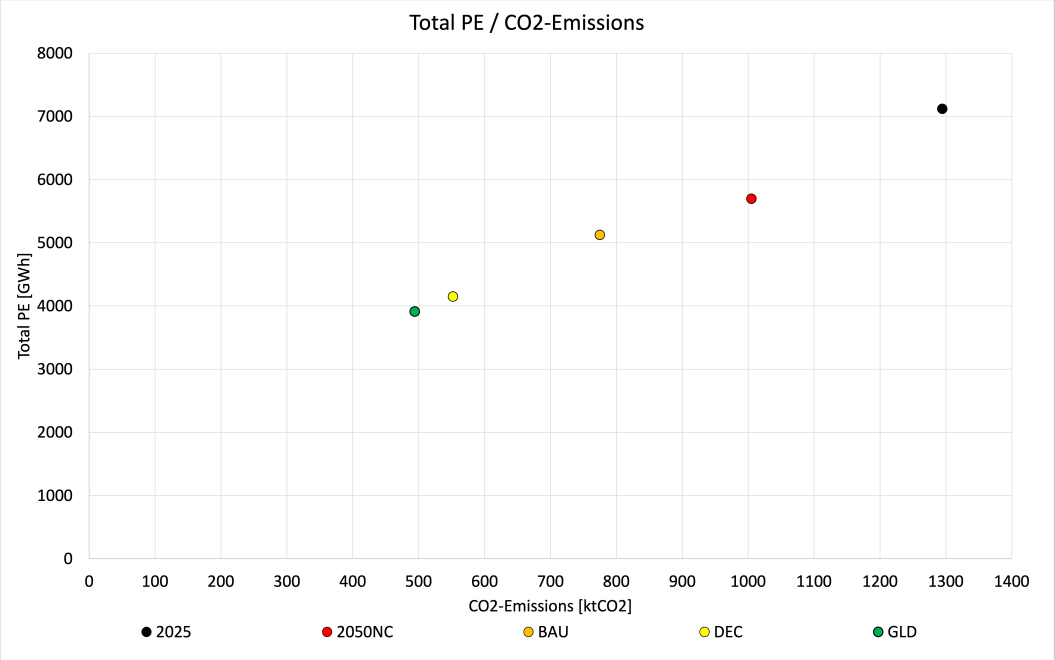

Figure 10: Emission and energy demand reduction pathway across scenarios.

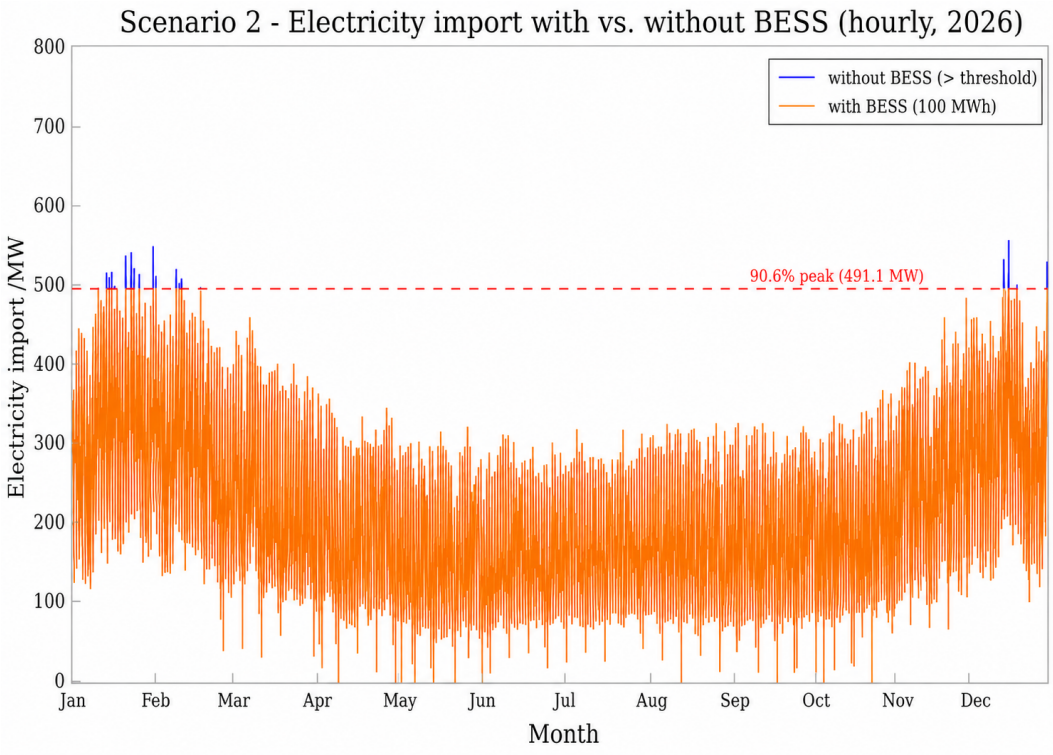

Figure 11: Effect of peak shaving with BESS for DEC.

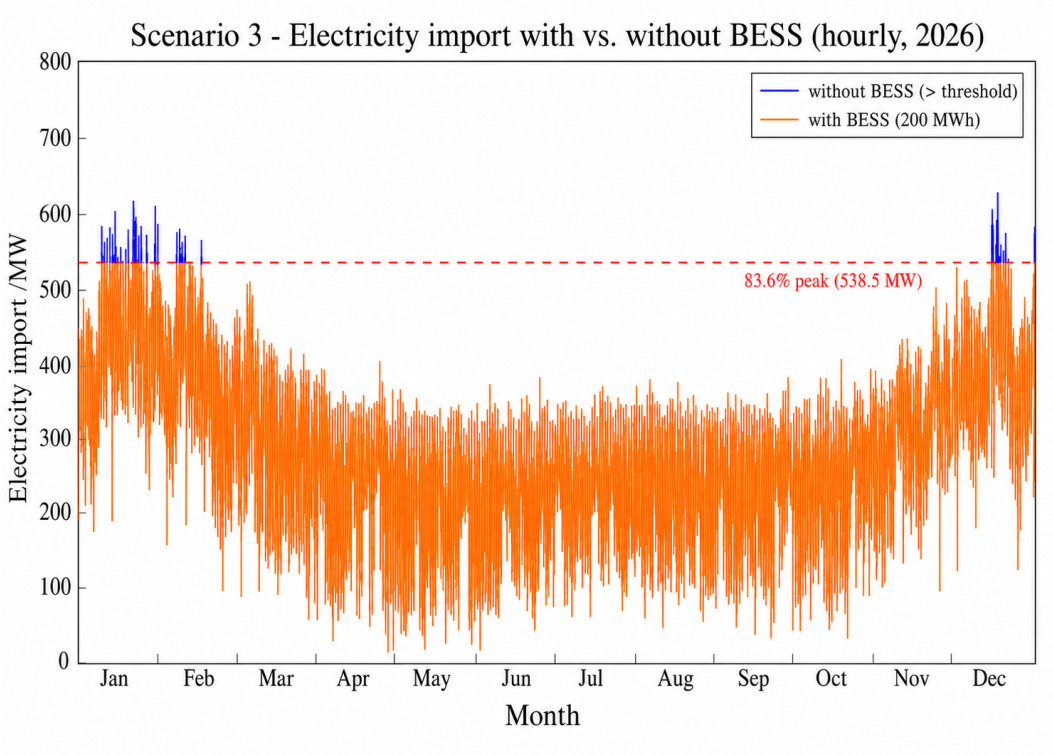

Figure 12: Effect of peak shaving with BESS for GLD.

4.3 System-Perspective Analysis

The system-perspective analysis evaluates the total economic and technical performance of the district energy system across the three transition scenarios. In contrast to the stakeholder perspective, this view considers the full system cost, including energy supply costs, technology-related costs, and system-wide effects such as private transport and petroleum consumption.

Table 14: Price assumptions used in the system-perspective economic analysis

Energy carrier / cost component	**Price**
Electricity	150 USD/MWh
District heat	100 USD/MWh
Petroleum	1.50 USD/liter
Natural gas	80 USD/MWh
CO_2 emissions	50 USD/tCO_2

The economic calculation is based on the assumed supply prices shown in Table 14. Electricity is valued at 150 USD/MWh, district heat at 100 USD/MWh, petroleum at 1.50 USD/liter, natural gas at 80 USD/MWh, and CO_2 emissions at 50 USD/tCO_2. These prices are used to calculate the system-level life-cycle cost (LCC) (see figure 13) of each scenario and compare it against the 2050NC reference case.

Table 15 presents the system perspective KPIs and indicators. The 2050NC reference case has an estimated LCC of 14.2 bn USD. Compared to this baseline, all three scenarios reduce total system costs. BAU reduces LCC by 15%, DEC by 23%, and GLD by 28%. GLD achieves the lowest total LCC with 10.2 bn USD, mainly due to higher electrification, higher renewable energy integration, and lower fossil fuel exposure.

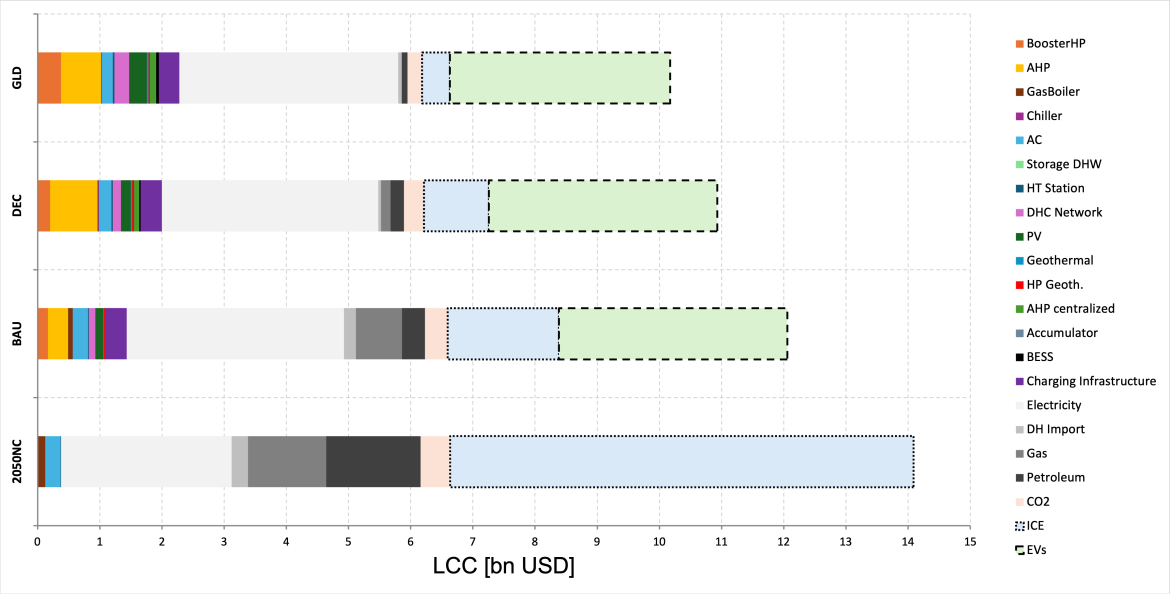

Figure 13: Life cycle cost of the portfolio from the system perspective.

At the same time, the technical performance improves across the scenarios. Total renewable energy share increases from 30% in BAU to 47% in GLD. Self-sufficiency rises from 10% to 20%, while electrification increases from 66% to 96%. This indicates that the more ambitious scenarios not only reduce emissions and fossil fuel dependency, but also improve the structural resilience of the district energy system.

Table 15 presents the system perspective KPIs and indicators. Life-cycle cost (LCC) reductions relative to the current state range from 15% under BAU to 28% under GLD. Total renewable energy share increases from 30% in BAU to 47% in GLD, with corresponding improvements in self-sufficiency (10% to 20%) and electrification rate (66% to 96%).

Table 16 presents the qualitative assessment across six system-level indicators. BAU scores low on system flexibility and backup capacity because it relies more strongly on conventional supply structures. DEC and GLD perform better due to higher technology diversification, increased storage and flexibility options, and stronger local value creation. GLD achieves the highest qualitative performance, particularly in system flexibility, backup capacities, local value creation, and employment impact.

Table 15: Scenario comparison: system-perspective economic and renewable energy indicators

KPI	2050NC	BAU	DEC	GLD
LCC [bn USD]	14.2	12.1	10.9	10.2
LCC reduction vs 2050 NoChanges	–	-15%	-23%	-28%
Total RES Share	15%	30%	39%	47%
Self-Sufficiency	0%	10%	12%	20%
Electrification	50%	66%	88%	96%

Table 16: Qualitative indicators by scenario

Indicator	BAU	DEC	GLD
System flexibility	low	high	very high
Technology diversification	medium	high	high
Backup capacities	low	high	very high
Price shock resilience	medium	high	high
Local value creation	medium	high	very high
Employment impact	medium	high	very high

4.4 Stakeholder-Perspective Economic Analysis

The stakeholder perspective evaluates the project from the viewpoint of actors that directly invest in, operate, and recover revenues from the energy system. In contrast to the system perspective, which assesses overall district-wide cost and performance, the stakeholder perspective focuses on economic feasibility as an implementable business case.

A technically optimal solution is not necessarily financially attractive for investors, operators, municipalities, or customers. The stakeholder view therefore considers whether the project generates sufficient revenues to recover capital expenditures, cover operational costs, and provide an acceptable return on investment. All system-related

costs are included, while private cars and petroleum costs are excluded (see Figure 14). Revenues are generated through customer payments based on retail prices for different end-use energy services.

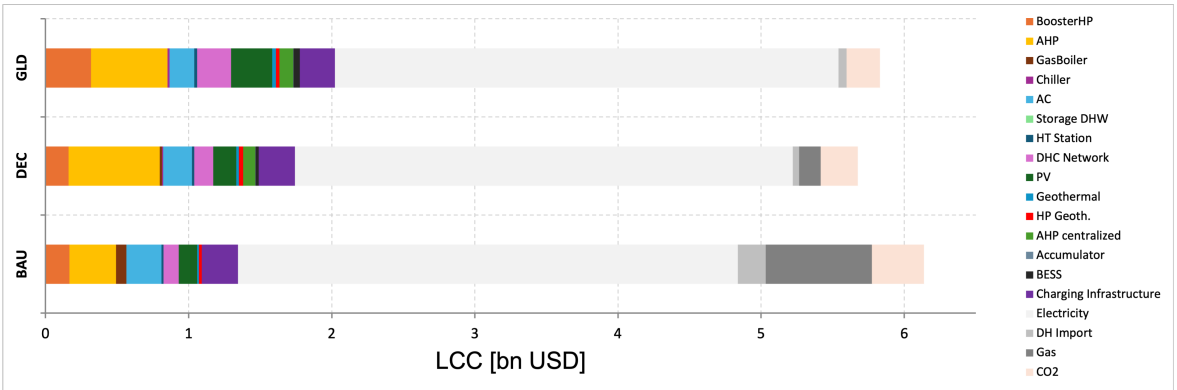

Figure 14: Life cycle cost of the portfolio from the stakeholder perspective.

Under this customer-revenue model, citizens pay regulated retail tariffs for useful energy services — heating, cooling, plug-load electricity, and e-mobility charging — rather than purchasing energy technologies directly. An operating entity finances, owns, and maintains all assets while customers pay for delivered services.

Retail prices are differentiated across scenarios to reflect differences in technology deployment, renewable integration, and service quality. BAU applies the lowest tariffs, reflecting its limited district energy integration and lower renewable share. DEC and GLD apply slightly higher tariffs, reflecting the additional value of increased electrification, higher renewable shares, and improved system flexibility — effectively a moderate green-service premium. Price increases are kept moderate to maintain social and political feasibility, with tariffs assumed to remain regulated and affordable rather than profit-maximising.

Table 17: Scenario-specific retail prices used for stakeholder revenue calculation

End-use	**BAU**	**DEC**	**GLD**
Heat [USD/kWh]	0.12	0.13	0.13
Cooling [USD/kWh]	0.13	0.14	0.14
Plug load [USD/kWh]	0.15	0.16	0.16
E-mobility [USD/kWh]	0.17	0.18	0.18

The tariffs in Table 17 generate the income streams used to evaluate NPV, IRR, BCR, and discounted payback period across scenarios.

Table 18: Stakeholder-perspective economic KPIs by scenario

KPI	BAU	DEC	GLD
NPV [million USD]	259	1,042	868
IRR [%]	7.6	12.2	10.2
BCR [-]	1.04	1.17	1.14
Discounted PB [years]	14	9	11
CAPEX [bn USD]	1.3	1.8	2.1

All three scenarios are economically feasible, each achieving a positive NPV and BCR above one, indicating that discounted benefits exceed discounted costs over the analysis period.

BAU achieves an NPV of 259 million USD, an IRR of 7.6 %, and a BCR of 1.04, with the lowest CAPEX of 1.3 bn USD. However, its discounted payback period of 14 years is the longest, reflecting slower capital recovery and weaker overall financial performance relative to the alternatives.

DEC delivers the strongest stakeholder economics: the highest NPV of 1,042 million USD, IRR of 12.2 %, and BCR of 1.17, with the shortest payback period of 9 years despite a CAPEX of 1.8 bn USD. The additional investment is compensated by stronger value creation from district energy integration, electrified heat supply, and local energy services.

GLD achieves an NPV of 868 million USD, an IRR of 10.2 %, and a BCR of 1.14, but requires the highest CAPEX of 2.1 bn USD, resulting in a payback period of 11 years. The scenario remains clearly profitable, though its higher investment intensity reduces financial attractiveness relative to DEC.

4.5 Sensitivity Analysis — District Energy Consideration

To assess the robustness of the economic case for the DEC scenario, a one-at-a-time sensitivity analysis was performed on four key parameters: the electricity price, the

discount rate, the seasonal coefficient of performance (COP) of the air-source heat pump (ASHP) system, and the ASHP capital cost. Rather than re-running the nPro tool for each parameter value, the baseline economic results of the DEC scenario were exported into an independent discounted-cash-flow (DCF) model, in which the four parameters were varied directly. This external approach allows each parameter to be swept continuously over a wide range at low computational cost. Because all sensitivity results are expressed as percentage deviations from the baseline, they are independent of the absolute NPV figure and therefore unaffected by differences between nPro's VDI 2067 method and the external DCF model.

Each parameter was swept in 10 % increments from −100 % to +100 % of its baseline value. Table 19 reports the tested ranges and the headline low/high scenarios discussed below. The extreme -100% cases are included only as mathematical boundary points for the sensitivity sweep and should not be interpreted as physically feasible technology assumptions.

Table 19: Tested parameter ranges for the Scenario DEC sensitivity sweep. Each parameter was varied from −100 % to +100 % of its baseline in 10 % increments. The baseline column corresponds to a 0 % change.

Change vs. base	**Electricity price ($/kWh)**	**Discount rate (%)**	**ASHP COP (–)**	**ASHP capital cost ($M)**
−100 %	0.000	0.0	0.00	0
−80 %	0.033	1.0	0.78	130
−60 %	0.066	2.0	1.55	260
−40 %	0.099	3.0	2.33	391
−20 %	0.132	4.0	3.10	521
0 % (base)	**0.164**	**5.0**	**3.88**	**651**
+20 %	0.197	6.0	4.66	781
+40 %	0.230	7.0	5.43	911
+60 %	0.263	8.0	6.21	1,041
+80 %	0.296	9.0	6.98	1,172
+100 %	0.329	10.0	7.76	1,302

The **discount rate** is the most influential parameter. Reducing it to 3 % (−40 %) raises the NPV by approximately 18 % ($+187 million), whereas increasing it to 10 % (+100 %) reduces it by approximately 33 % ($-344 million), yielding a total scenario range of $531 million. This asymmetry reflects the long investment horizon and large upfront capital outlay, which amplify the compounding effect at higher discount rates.

The **electricity price** is the second-most sensitive parameter because it affects both grid-electricity purchases and revenues from plug loads and e-mobility. The relationship is approximately linear: each 1 % change in electricity price produces a ±0.487 % change in NPV. A low-price scenario of $0.115 / kWh (−30 %) reduces the NPV by roughly 15 % ($-156 million), while a high-price scenario of $0.253 / kWh (+54 %) increases it by approximately 24 % ($+250 million), giving a total range of $406 million.

The **ASHP capital cost** exhibits a linear and symmetric sensitivity, with each 1 % variation producing a ±0.161 % shift in NPV. A −50 % cost reduction increases the NPV by approximately 8 % ($+83 million), while a +50 % cost increase reduces it by the same amount, giving a total range of $166 million.

Finally, the **ASHP COP** displays a nonlinear response due to its inverse relationship with electricity consumption: $Q_{\text{elec}} \propto 1/\text{COP}$. A downside scenario of COP 2.33 (−40 %) reduces the NPV by roughly 4 % ($-42 million), while an upside of COP 4.50 (+16 %) increases it by approximately 1 % ($+10 million), giving a total range of $52 million. The response becomes strongly asymmetric at very low COP values, indicating that the ASHP system must maintain sufficient seasonal performance for the project economics to remain viable.

Spider chart — normalised parameter elasticities

Figure 15 presents the spider chart, which plots the percentage impact on NPV against the percentage change in each parameter over the full −100 % to +100 % sweep of Table 19. By normalising every parameter onto a common horizontal axis, the chart isolates the per-unit-change elasticity of each driver. The discount rate and electricity price exhibit the steepest slopes and therefore the largest elasticities, while the ASHP capital cost shows a more moderate linear influence. The ASHP COP is visibly nonlinear and bends sharply downward once the COP falls below roughly 2.5, creating the "COP cliff" shown in the figure. The spider chart confirms the parameter ranking: discount rate, electricity price, ASHP capital cost, and ASHP COP.

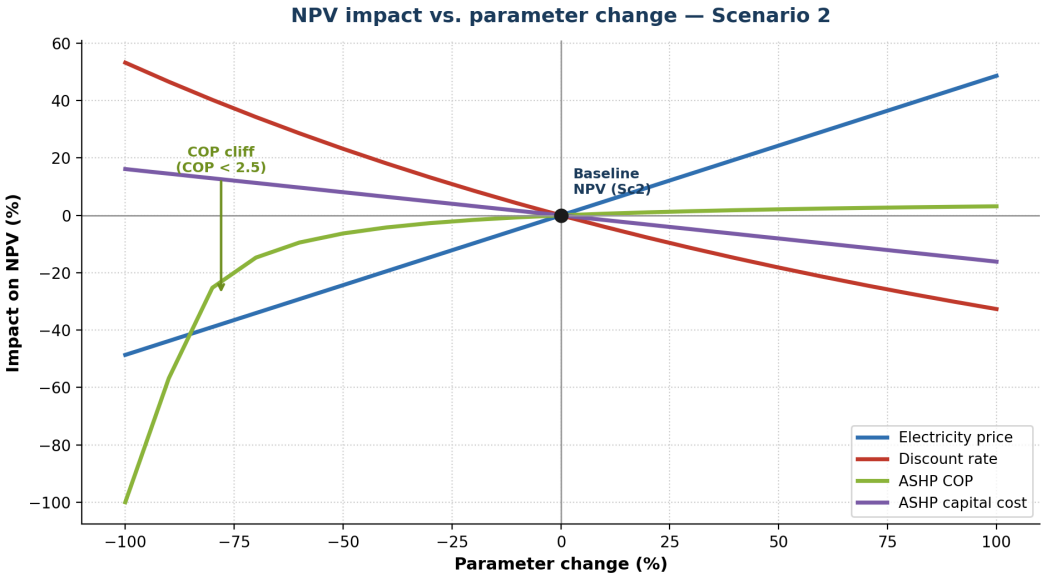

Figure 15: Spider chart of NPV sensitivity for Scenario DEC.

Tornado Diagram — DEC Scenario-based NPV Ranges

Figure 16 presents the tornado diagram, which ranks the four parameters by their absolute NPV impact relative to the baseline of $1,042 M. The diagram confirms the same ranking observed in the spider chart: discount rate ($-344 M to $+187 M), electricity price ($-156 M to $+250 M), ASHP capital cost ($83 M either side), and ASHP COP ($-42 M to $+10 M). The asymmetry of the discount-rate and electricity-price ranges reflects the nonlinear discounting behaviour and the asymmetric market-price scenarios.

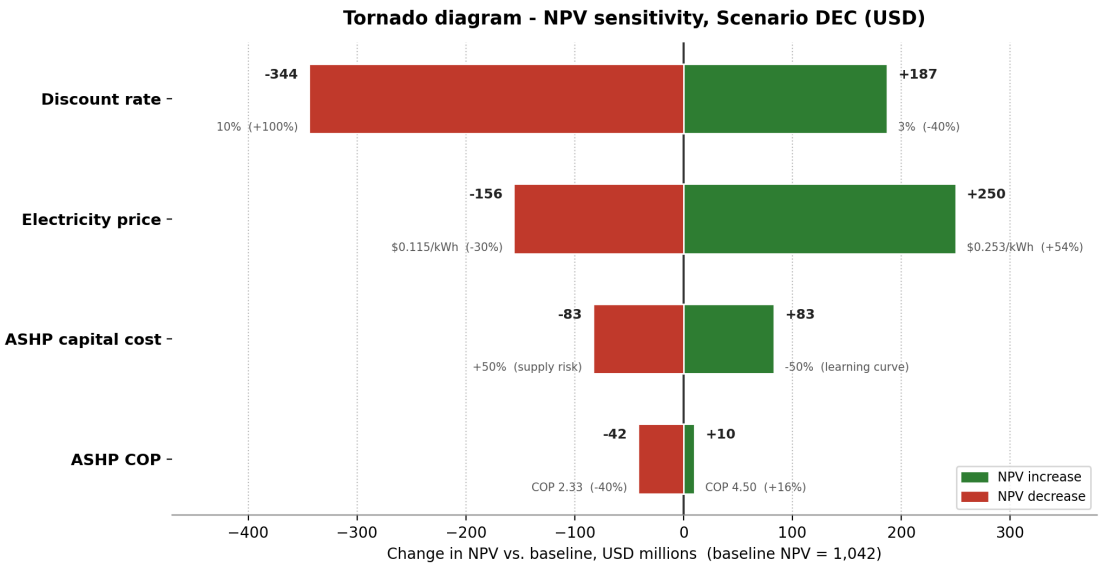

Figure 16: Tornado diagram for Scenario DEC, illustrating the relative NPV range associated with each sensitivity parameter.

Overall, the tornado diagram ranks the four parameters in descending order of impact as: (1) discount rate, (2) electricity price, (3) ASHP capital cost, and (4) ASHP COP.

The ranking reflects both the parameter elasticity and the chosen scenario widths. In all tested scenarios, the NPV of Scenario DEC remains positive, indicating that the District Energy First pathway is economically robust across a wide range of plausible future conditions.

4.6 Trade-off interpretation

The scenario comparison highlights a clear trade-off between environmental ambition, economic performance, and implementation feasibility. BAU requires the lowest investment and therefore represents the least complex transition pathway, but it also delivers the weakest reductions in CO_2 emissions, primary energy demand, and peak load. GLD achieves the strongest environmental performance because it combines full fossil heating phase-out, higher renewable integration, and larger storage capacities. However, these benefits come with higher CAPEX, stronger reliance on renewable capacity outside the district boundary, greater infrastructure complexity, and a longer payback period.

DEC provides the most balanced outcome across the three KPIs. It achieves sub-

stantial emissions and primary energy reductions while also improving peak-load management through coordinated EV charging, battery storage, and district energy integration. At the same time, it avoids the highest investment and implementation risks of GLD and delivers the strongest stakeholder-economic performance. The main trade-off is therefore not between action and inaction, but between maximum technical decarbonisation and a more robust pathway that can realistically be financed, governed, and implemented. From this perspective, DEC represents the preferred compromise between environmental impact, economic attractiveness, and practical feasibility.

5 Business Model Innovation

The proposed business model, *Gangdong Energy Hub*, is structured as a two-tier public-private Energy-as-a-Service (EaaS) platform designed to facilitate the district's transition towards a low-emission and highly electrified urban energy system. The model directly addresses one of the primary barriers to urban energy transition: the fragmentation of energy infrastructure and the prohibitive upfront investment required from individual building owners.

5.1 Two-Tier Capital Structure

The Energy Hub is organised into two complementary tiers that together deliver one integrated service across all considered scenarios. The overall architecture, including the two capital tiers, the integrated service platform, and the flow of services and tariff revenue, is shown in Figure 17.

The **public tier** covers approximately 25–35% of total capital expenditure (CAPEX) and is responsible for the strategic backbone infrastructure. This includes the 5th-generation district heating and cooling (5GDHC) network, central thermal storage, central heat pumps, and grid modernisation assets. These are long-lived assets with broad public value that are not always sufficiently attractive for private investors acting alone. Public stakeholders, including Gangdong-gu, the Seoul Metropolitan Government, KDHC, and KEPCO, co-own this layer and take on the associated long-term infrastructure risk.

The **private tier** covers approximately 65–75% of CAPEX and encompasses customer-

facing assets operated by a Special Purpose Vehicle (SPV). Equity within the SPV is distributed according to each operator's CAPEX contribution: the building and HVAC operator holds approximately 75%, the PV operator approximately 15%, the EV charging operator approximately 7%, and the battery and flexibility operator approximately 3%. This structure ensures that investment exposure and financial returns are proportionally aligned, avoiding arrangements in which one partner bears a disproportionate share of capital while receiving equivalent returns to partners with smaller contributions.

The CAPEX split between tiers shifts by approximately 10 percentage points across the three scenarios considered, but the underlying architecture remains unchanged. This scenario-robustness is a key feature of the model, as it avoids the need for a fundamentally different governance or ownership structure depending on the policy pathway followed.

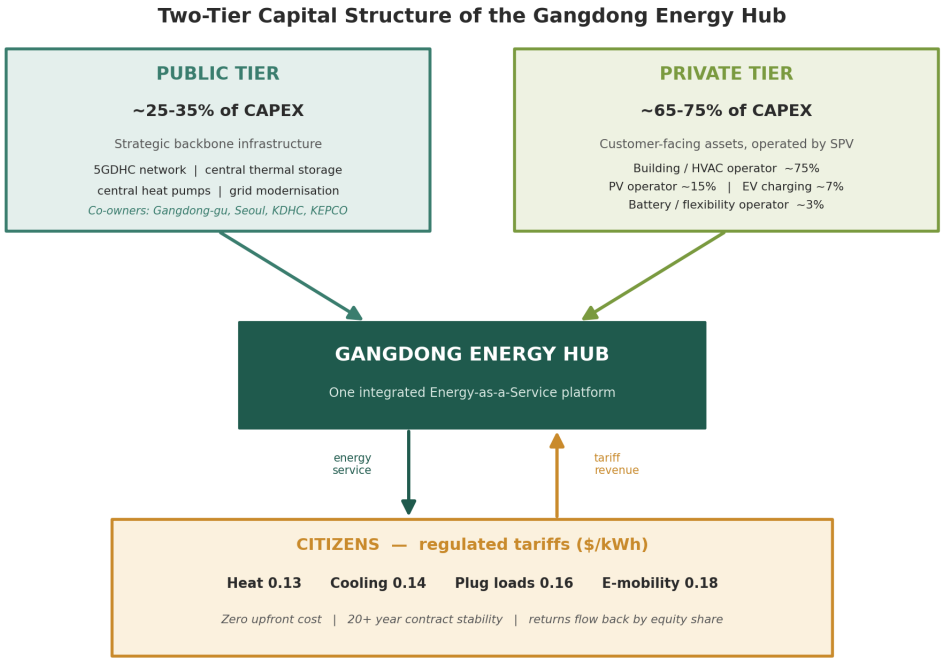

Figure 17: Two-tier capital structure of the Gangdong Energy Hub.

5.2 Service Model and Revenue Streams

Rather than requiring customers to purchase and operate energy technologies independently, the Energy Hub finances, owns, and maintains all assets while customers pay

regulated tariffs for delivered energy services. Citizens subscribe to a service rather than invest in equipment. Indicative regulated tariffs are set at \$0.13/kWh for heat, \$0.14/kWh for cooling, \$0.16/kWh for plug loads, and \$0.18/kWh for e-mobility services, with contract stability guaranteed over a minimum 20-year horizon. Returns flow back to investors and operators according to their respective equity shares.

Revenue is diversified across four main streams: heat services (approximately 35%), plug loads (approximately 40%), cooling (approximately 12%), and e-mobility (approximately 13%). This diversification reduces exposure to fluctuations in any single market segment and improves the financial resilience of the platform.

The model integrates services that are typically managed in isolation, including electricity supply, heating, cooling, EV charging, and building-level demand management, into one coordinated district-scale platform. Innovation lies not only in the individual technologies deployed, but in the aggregation, ownership structure, operational coordination, and the shift from individual building optimisation to district-scale optimisation. Buildings with complementary demand profiles, such as residential evening peaks, commercial daytime loads, and commercial refrigeration waste heat, can be coordinated to achieve substantially higher efficiency than isolated systems.

5.3 Customer Segments and Value Proposition

The primary customer segments are apartment complexes, public buildings, schools, commercial facilities, and small businesses. Apartment associations are particularly significant because they aggregate large numbers of households under long-term contracts. Public buildings serve as anchor customers that reduce investment risk during early implementation phases and provide visible demonstration of technical feasibility. Commercial customers with simultaneous heating, cooling, and refrigeration demands create opportunities for waste heat recovery and load balancing.

For residents, the model eliminates upfront capital requirements and technical responsibility. Benefits include lower and more stable monthly energy bills, reliable heating and cooling, improved access to EV charging infrastructure, and reduced maintenance burden. Apartment associations and building owners receive improved energy performance, infrastructure modernisation, and increased long-term property value without carrying the full investment burden. For the municipality, the model provides

a practical mechanism for reducing emissions and peak electricity demand in alignment with local climate targets.

5.4 Citizen Acceptance and Governance

The transition towards electrified district energy will occur regardless of the specific implementation model, given Korea's binding net-zero 2050 commitment. The relevant question is therefore whether electrification proceeds building by building, with high upfront costs, duplicated infrastructure, and uncoordinated investment, or through a district-scale platform that reduces total system cost and improves coordination. Coordinated district-scale implementation is expected to be more cost-effective than fragmented building-level approaches.

Beyond cost efficiency, the model is designed to strengthen rather than reduce citizen agency. Under existing arrangements, energy decisions are largely determined by apartment associations and grid utilities, with limited direct input from individual residents. The proposed governance structure extends participation through citizen seats on the SPV board and public golden-share rights, providing residents with formal mechanisms to contest tariff increases and operational decisions they consider unreasonable.

Financial protection is ensured through three structural commitments. First, the model is **auditable**: full costs, operator margins, and carbon revenues are published annually through open-book accounting, ensuring that citizens can verify how their payments are used. Second, the model is **vetoable**: public golden-share rights and citizen representation on the SPV board provide a formal check against unjustified tariff increases or decisions contrary to the public interest. Third, the model is **exitable**: participants retain the right to renegotiate or exit the service agreement after year seven, preventing permanent contractual lock-in.

The governance case for the Energy Hub is therefore not an appeal to trust in any specific private operator. Instead, it is an appeal to trust in a governance structure that is auditable, vetoable, and exitable, and that provides citizens with more formal rights and protections than the fragmented arrangements it replaces. Figure 18 summarises this argument, pairing the value delivered to citizens with the structural safeguards that keep that value protected over the contract horizon.

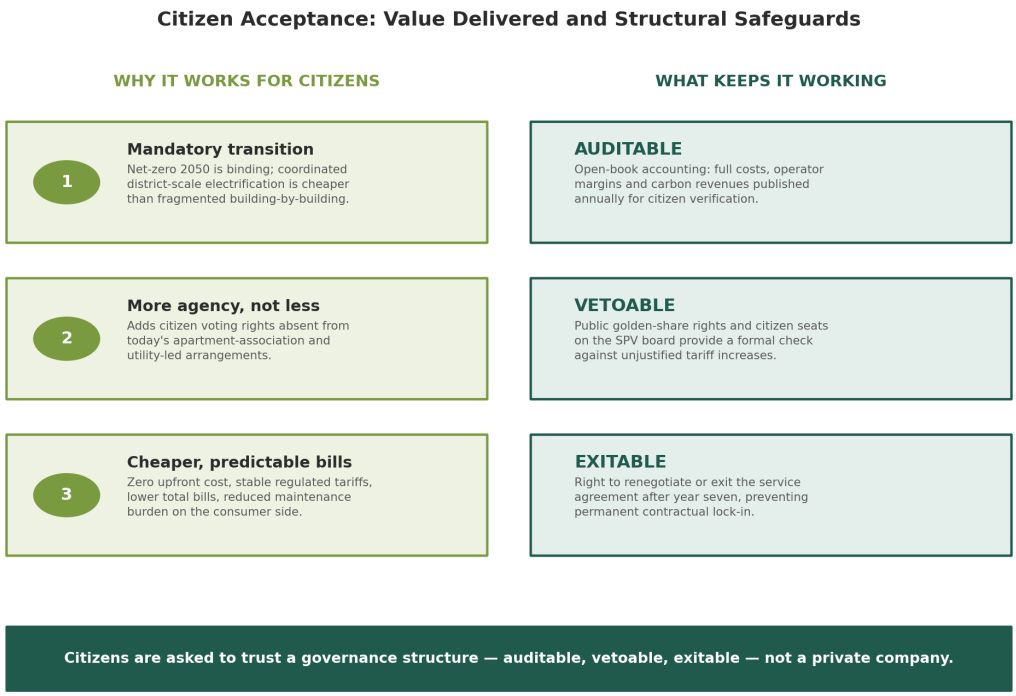

Figure 18: Citizen acceptance logic of the Gangdong Energy Hub.

5.5 Implementation Strategy and Risk Mitigation

Implementation follows a phased rollout to manage investment risk and validate technical and financial performance prior to full-scale expansion. The initial phase establishes a demonstration cluster comprising several apartment buildings, a public anchor building, rooftop PV installations, smart energy management systems, EV charging infrastructure, and a small low-temperature heat-sharing network. Subsequent phases expand to neighbouring building clusters with additional renewable generation, storage capacity, and broader demand-response coordination, ultimately delivering a fully integrated district energy platform across Gangdong-gu.

Principal risks include high upfront capital requirements, customer acquisition uncertainty, technical integration complexity, and regulatory uncertainty regarding local energy sharing and flexibility markets. These risks are mitigated through phased deployment, long-term service contracts, the use of public buildings as anchor customers, reliance on proven technologies in early stages, and close collaboration with licensed utilities where required.

A further risk specific to this model is multi-stakeholder coordination. The two-tier ownership boundary, proportional equity allocation, and the golden-share and open-

book accounting mechanisms provide a pre-agreed framework for resolving disputes, while phased deployment allows the coordination model to be tested at demonstration scale before district-wide rollout.

From an investor perspective, the combination of long-term contracted revenue, diversified income streams, and a scalable model replicable across other Seoul districts provides infrastructure-like cash flow characteristics attractive relative to short-term energy market volatility.

6 Discussion and Limitations

6.1 Reduced Representation of the Building Stock in nPro

The real building stock of Gangdong-gu comprises 20,192 main buildings, based on 2018 building registry data [44]. In contrast, the nPro model represents the district through fewer than 50 aggregated demand points. This aggregation was necessary to keep the model computationally manageable, but it limits the spatial accuracy of the district heating and cooling network representation.

As a result, network length, building connections, and distribution costs cannot be directly derived from the nPro model. Therefore, DHC network costs were estimated externally using simplified assumptions for connected buildings, average pipe length, routing factors, pipe costs, installation costs, and connection costs.

6.2 Uncertainty of Input Assumptions

The scenario results are subject to uncertainty in several key input assumptions. Technology costs, electricity prices, grid emission factors, discount rates, implementation rates, and DHC network costs can vary substantially over the 2035–2050 study period. In particular, the economic results are sensitive to assumptions regarding CAPEX, electricity tariffs, and discount rate, while the environmental results depend strongly on the future carbon intensity of the public electricity grid. The results should therefore be interpreted as scenario-based estimates rather than precise forecasts.

6.3 Electricity Price Reduction due to Peak Shaving

Peak shaving was adopted as a key performance indicator to capture the system's ability to reduce maximum electricity demand from the grid. However, the modelling framework used in this study treats the electricity price as a static input parameter, meaning that reductions in peak demand are not reflected in a corresponding reduction in electricity cost within the model. In practice, peak shaving can lower electricity expenditure through reduced capacity charges and more favourable grid tariffs, but this feedback mechanism is absent from the current implementation.

This represents a limitation of the study, as the full economic benefit of peak shaving is likely underestimated in the cost optimisation results. Future work could address this by adopting a dynamic or time-of-use electricity pricing structure, where grid tariffs vary with demand levels, allowing the model to capture the cost savings associated with reduced peak loads. Incorporating such a pricing model would strengthen the link between the peak shaving KPI and the economic objective function, and would likely increase the relative attractiveness of storage technologies that contribute to peak reduction.

6.4 Stochastic Peak Loads and Peak Shaving Capacity

Another limitation concerns the treatment of stochastic peak loads and peak-shaving capacity. In reality, annual peak loads are not fully deterministic, as they depend on weather conditions, occupancy patterns, user behaviour, operational schedules, and other uncertain factors. Therefore, the peak load can vary between years even if the underlying energy system and demand structure remain unchanged.

In this study, the stochastic peak load was quantified based on the available load profile and used as a representative value for the analysis. However, this peak should not be interpreted as a fixed value that will occur identically in every year of the project lifetime. Instead, it represents a probable peak condition derived from the modelled demand profile.

The peak-shaving capacity was treated differently. It was defined as an absolute energy capacity in MWh and therefore kept constant across the analysis. This means that the technical ability of the system to shift or reduce peak demand was assumed to remain fixed, while the actual occurrence and magnitude of peak loads remains

probabilistic.

Conceptually, this can be described as:

$$E_{\text{shaving}} = \text{constant} \tag{6}$$

while the annual peak load is represented as a stochastic variable:

$$P_{\text{peak},y} = f(\text{weather}_y, \text{occupancy}_y, \text{operation}_y, \text{demand}_y) \tag{7}$$

where $P_{\text{peak},y}$ is the peak load in year y. The peak-shaving capacity can therefore reduce or shift a fixed amount of energy, but the relative impact of this capacity may differ between years depending on the realised peak load.

The relative peak-shaving effect can be expressed as:

$$r_{\text{shaving},y} = \frac{E_{\text{shaving}}}{E_{\text{peak},y}} \tag{8}$$

where $E_{\text{peak},y}$ represents the energy associated with the peak period in year y. Since E_{shaving} is constant but $E_{\text{peak},y}$ may vary, the relative contribution of peak shaving is also year-dependent.

As a result, the reported peak-shaving effect should be interpreted as an indicative value based on the quantified stochastic peak condition. It does not guarantee that the same percentage of peak reduction will be achieved in every year. Instead, the absolute peak-shaving capacity remains constant, while the load situation to which it is applied is subject to uncertainty.

6.5 Transport Assumptions

The transition of the private car fleet and the overall reduction in car usage are two interdependent levers that strongly influence the energy demand of the transport sector. Both are subject to policy decisions at the local and national government level, and their trajectories differ substantially across the three scenarios considered in this study.

6.5.1 Electric Vehicle Share

The electrification of the private car fleet is driven primarily by regulatory mandates rather than market forces alone. The BAU scenario follows South Korea's Nationally

Determined Contribution [43], which targets 70% of all new vehicles sold in 2035 to be electric. The moderate and ambitious scenarios assume progressively stronger legislative intervention, analogous to measures such as the European Union's effective ban on new internal combustion engine sales from 2035, accelerating fleet turnover beyond the NDC baseline.

6.5.2 Private Car Reduction

A reduction in the total number of private car trips is essential for managing peak energy demand and achieving broader sustainability goals. This reduction depends critically on governmental action on two fronts. First, investment in and improvement of public transport has been shown to reduce car ownership and usage: a study of the Copenhagen metropolitan area found that a metro network extension alone reduced car ownership by 2–3%, with car ownership consistently lower in areas well-served by public transit [28]. A broader literature review further confirms that when public transport investment successfully shifts travel behaviour, it reduces road congestion, energy use, and emissions [3]. Second, demand-side regulation (such as congestion charges) can directly suppress private car use. Stockholm's *trängselskatt* provides a well-documented example: introduced permanently in 2007, it reduced car traffic into the inner city by 20% during rush hours, and over the subsequent decade Stockholm's population grew by 10% while city-wide traffic levels fell by 22% [34]. The degree to which such instruments are deployed by local government determines the assumed car reduction level across the three scenarios.

6.6 Cost Estimation of the DHC Network

The DHC network cost estimation was carried out as an external post-processing step, since nPro represents Gangdong-gu with aggregated demand points and therefore does not capture the actual number of building connections. In reality, Gangdong-gu comprises 20,192 main buildings based on 2018 building registry data [44]. Hence, an order-of-magnitude estimate was used to approximate the required network CAPEX.

The number of connected buildings, $N_{\text{connected}}$, was calculated from the total number of buildings, N_{total}, and the scenario-specific implementation ratio, $r_{\text{implementation}}$:

$$N_{\text{connected}} = N_{\text{total}} \cdot r_{\text{implementation}} \tag{9}$$

Assuming that one pipe segment connects two opposite buildings along a street, the number of pipe segments, N_{pipe}, was estimated as:

$$N_{\text{pipe}} = \frac{N_{\text{connected}}}{2} \tag{10}$$

The total pipe length, L_{pipe}, was then calculated using an average building distance of $d_{\text{avg}} = 40\,\text{m}$ and an increasing factor, f_{inc}, to account for routing deviations and network complexity:

$$L_{\text{pipe}} = N_{\text{pipe}} \cdot d_{\text{avg}} \cdot f_{\text{inc}} \tag{11}$$

Pipe costs, C_{pipe}, were calculated from the total pipe length and the sum of pipe material costs, c_{pipe}, and installation costs, c_{install}, per metre:

$$C_{\text{pipe}} = L_{\text{pipe}} \cdot (c_{\text{pipe}} + c_{\text{install}}) \tag{12}$$

Building connection costs, $C_{\text{connection}}$, were estimated from the number of connected buildings and the average connection cost per building, $c_{\text{connection}}$:

$$C_{\text{connection}} = N_{\text{connected}} \cdot c_{\text{connection}} \tag{13}$$

The total DHC network CAPEX, $C_{\text{DHC,network}}$, was obtained as:

$$C_{\text{DHC,network}} = C_{\text{pipe}} + C_{\text{connection}} \tag{14}$$

This approach provides an indicative order-of-magnitude estimate only. It does not include detailed geospatial routing, hydraulic optimisation, underground infrastructure constraints, or building-specific connection requirements.

6.7 The Exclusion of Waste Management

The waste sector, currently responsible for 11% of CO_2eq emissions, is largely excluded from this report. Biofuel-derived energy potential was deemed insufficient for inclusion, as discussed in Section 4.2.4. Waste incineration was similarly excluded due to documented political opposition in Seoul [25, 45].

Several waste reduction measures are already in effect or planned for Gangdong-gu. A metropolitan-wide landfill ban, effective 2026, is expected to curb emissions. At the district level, an RFID-based pay-as-you-throw system for food waste has been deployed in new residential developments [11]. Additionally, expanded recycling infrastructure, including dedicated vinyl collection and improved waste intake management, has been introduced to reduce reliance on landfill and incineration [7]. Based on a seven-year average, the district projects a 10.6% reduction in waste volume by 2026.

These measures are expected to yield a net reduction in waste-sector CO_2eq emissions. However, insufficient data on the long-term impact of the landfill ban and complementary initiatives precluded a 2050 emissions projection; this remains an area for future work.

7 Conclusion, Policy and Strategy Recommendations

All three transition scenarios confirm that Gangdong-gu can achieve substantial reductions in energy demand, carbon emissions, and grid peak load by 2050, and all remain economically viable. The question is therefore not whether to act, but which pathway best balances environmental performance, economic viability, and implementation realism.

DEC is recommended as the primary implementation pathway. It delivers a 57% reduction in CO_2 emissions, a 42% reduction in primary energy demand, and an 14% normalised peak-shaving effect, while achieving the strongest stakeholder economics (NPV of 1,042 million USD, IRR of 12.2%, payback of 9 years). It captures most of GLD's environmental benefits while avoiding its higher capital requirements, external renewable dependence, and implementation complexity. GLD remains a viable high-ambition alternative if storage costs decline, grid decarbonisation accelerates, or stricter climate targets demand deeper reductions.

The immediate priority is establishing governance before scaling technology. The proposed *Gangdong Energy Hub* — a public-private Energy-as-a-Service platform with a two-tier capital structure — provides the institutional foundation, but requires early engagement with Gangdong-gu, the Seoul Metropolitan Government, KEPCO, KDHC, private operators, and building associations. Implementation should begin with a phased demonstration cluster comprising apartment buildings, a public anchor building, rooftop PV, coordinated EV charging, thermal storage, and a small 5GDHC network, before expanding district-wide.

Finally, results should be read as comparative scenario outcomes rather than exact forecasts. Absolute values may shift with updated building data, technology costs, or grid assumptions, but the relative ranking of pathways remains robust across the consistent modelling framework applied throughout.

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A Additional Figures

Figure 19: nPro modelling approach for the BAU scenario.

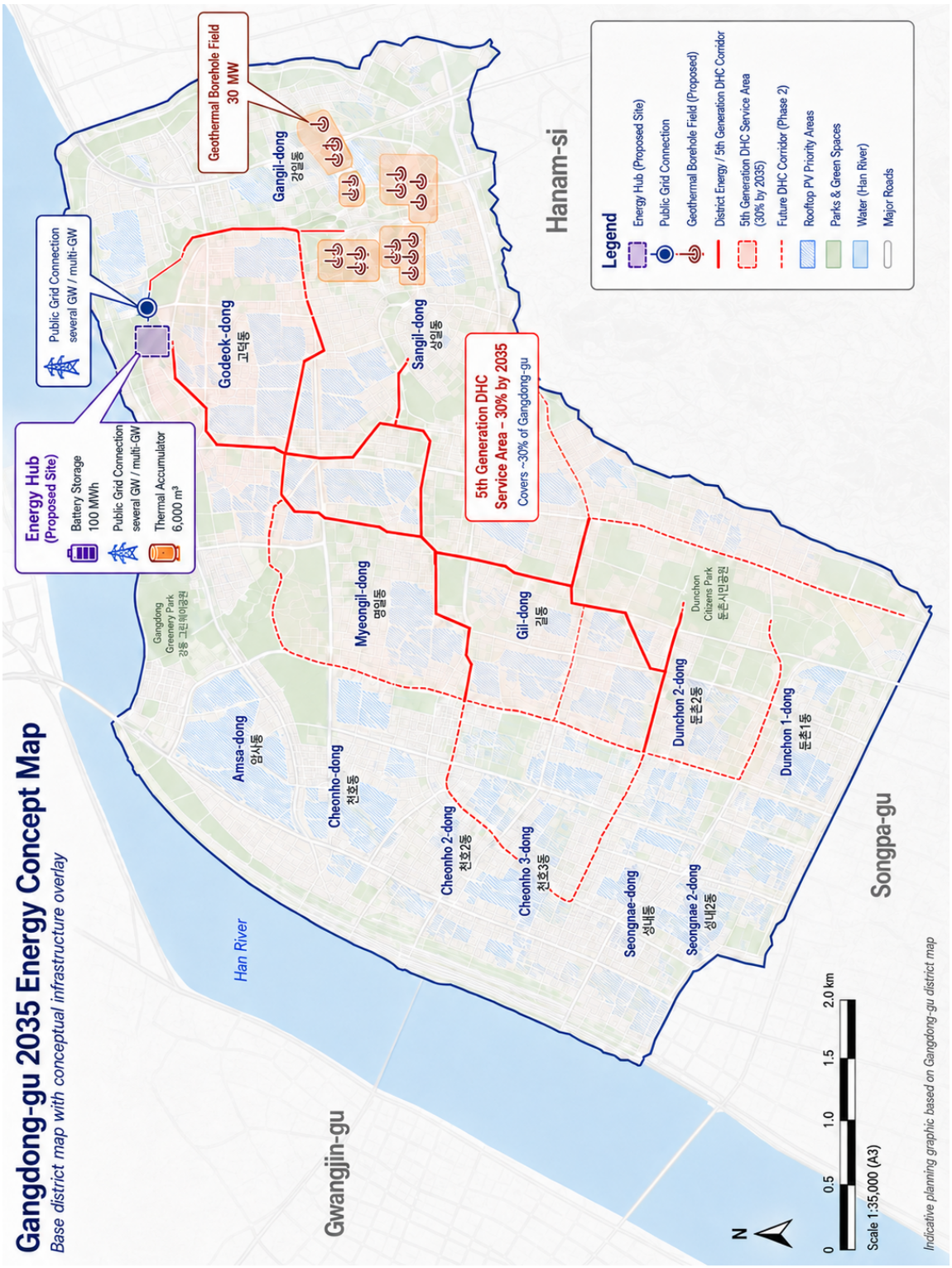

Figure 20: Gangdong-gu 2035 energy concept map for the DEC scenario.

Figure 21: Point-of-view illustration of the Gangdong-gu 2035 energy hub concept, created with AI.

Figure 22: Point-of-view illustration of the Gangdong-gu 2035 geothermal concept, created with AI.

B South Korea Data Figures

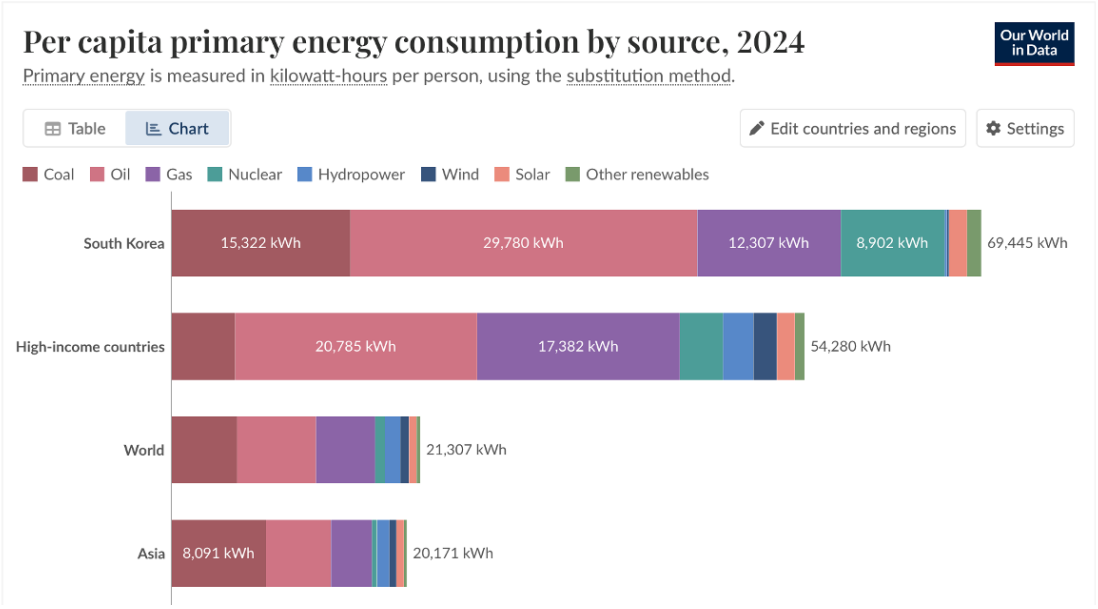

Figure 23: Per-capita primary energy consumption by source in 2024, comparing South Korea with high-income countries, the world average, and Asia. Source: Our World in Data.

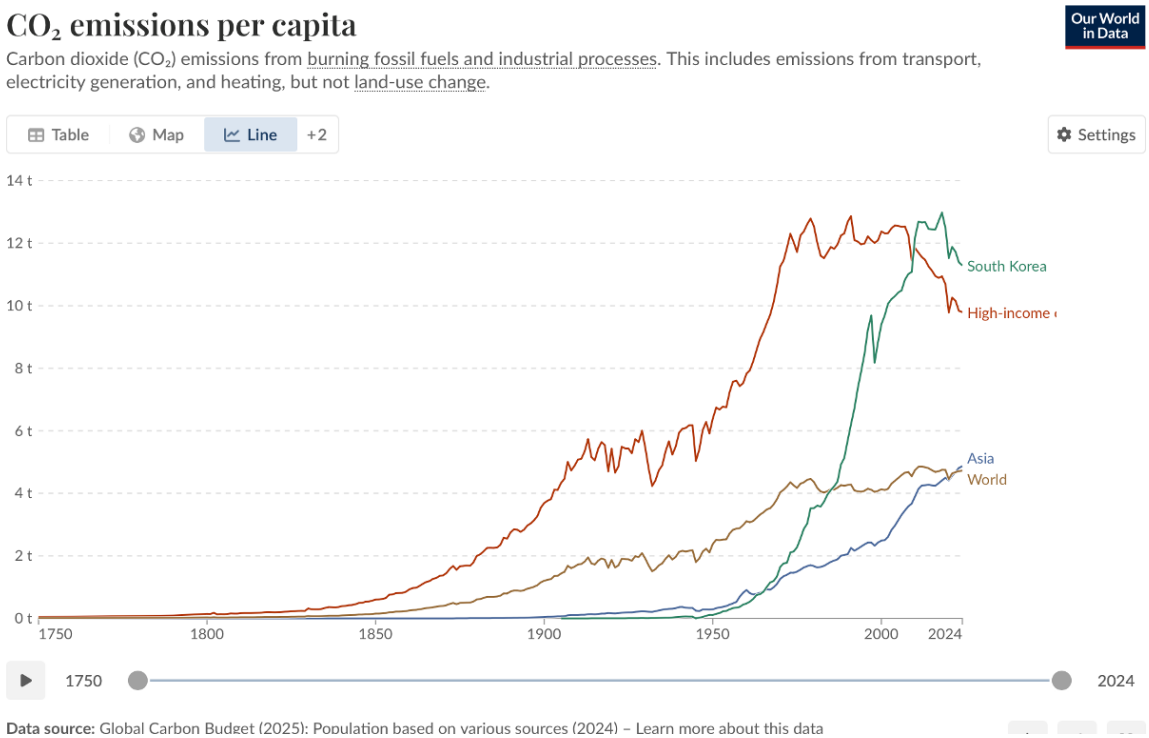

Figure 24: Historical CO_2 emissions per capita for South Korea, high-income countries, Asia, and the world average. Source: Our World in Data.

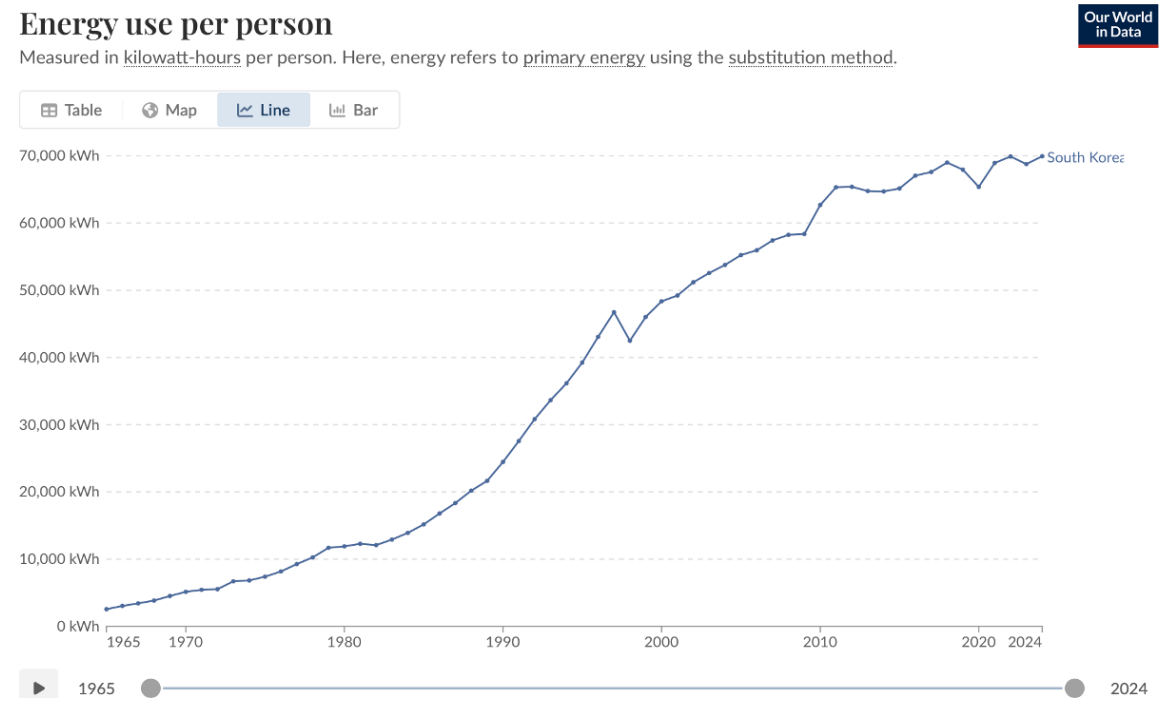

Figure 25: Historical development of primary energy use per person in South Korea. Source: Our World in Data.

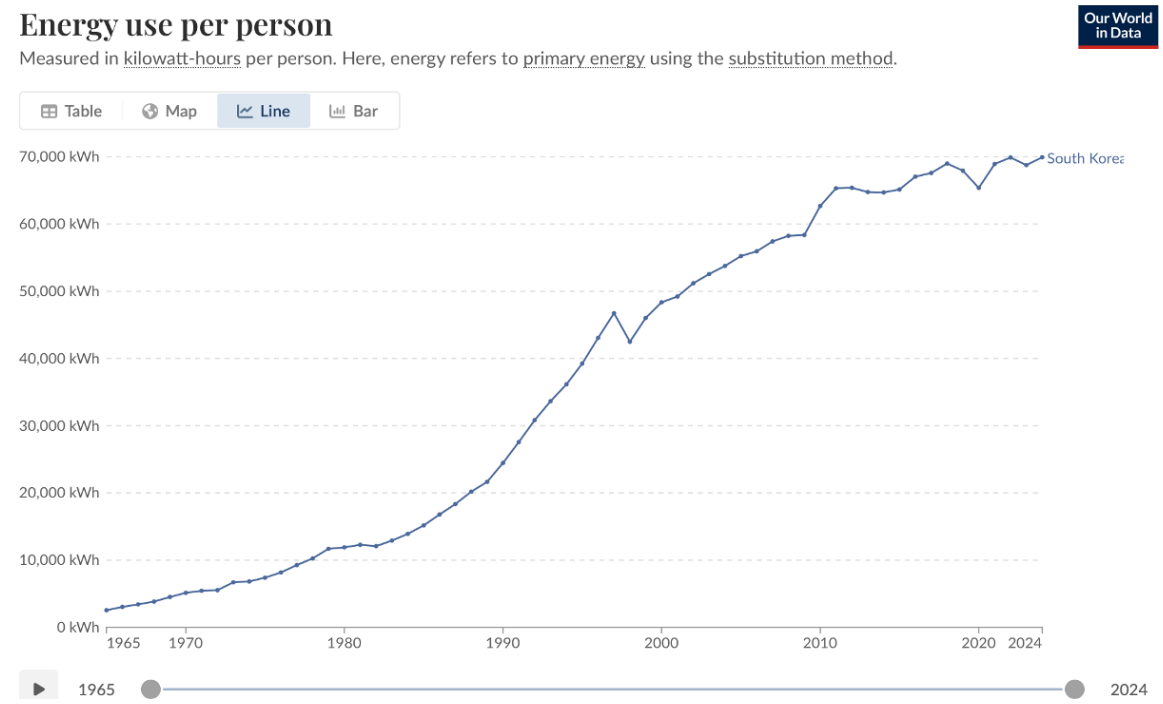

Figure 26: Primary energy use per person in South Korea, measured in kilowatt-hours per capita using the substitution method. Source: Our World in Data.

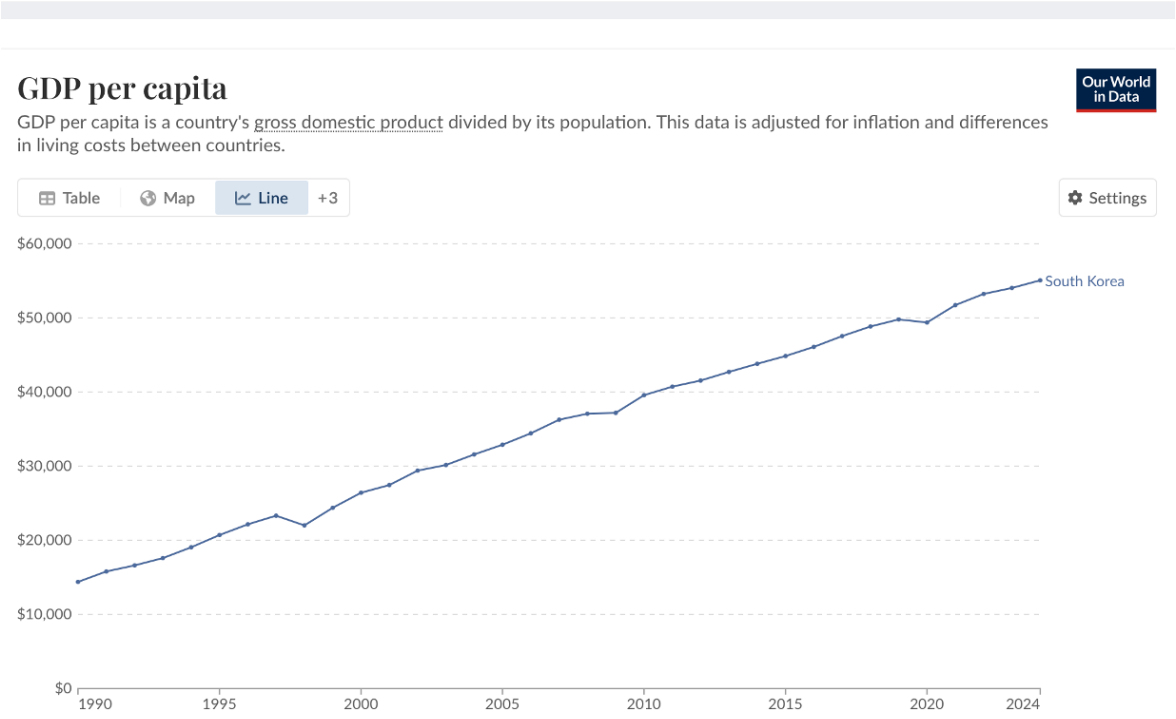

Figure 27: Historical development of GDP per capita in South Korea, adjusted for inflation and differences in living costs. Source: Our World in Data.

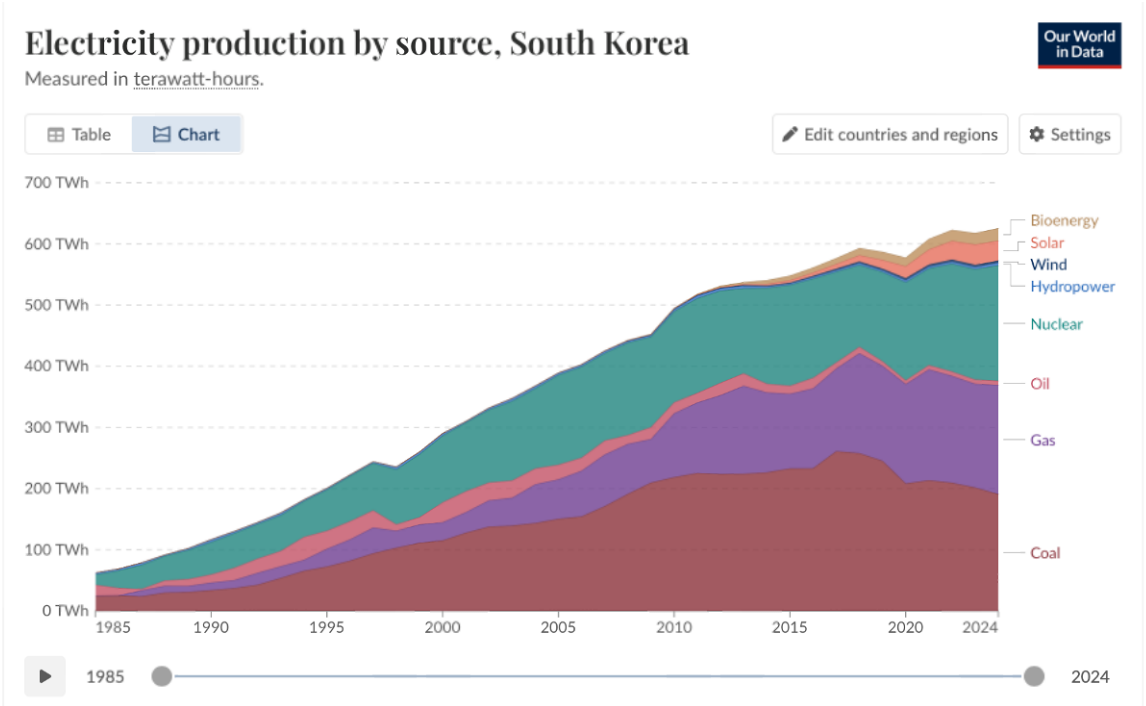

Figure 28: Electricity production by source in South Korea, measured in terawatt-hours. Source: Our World in Data.

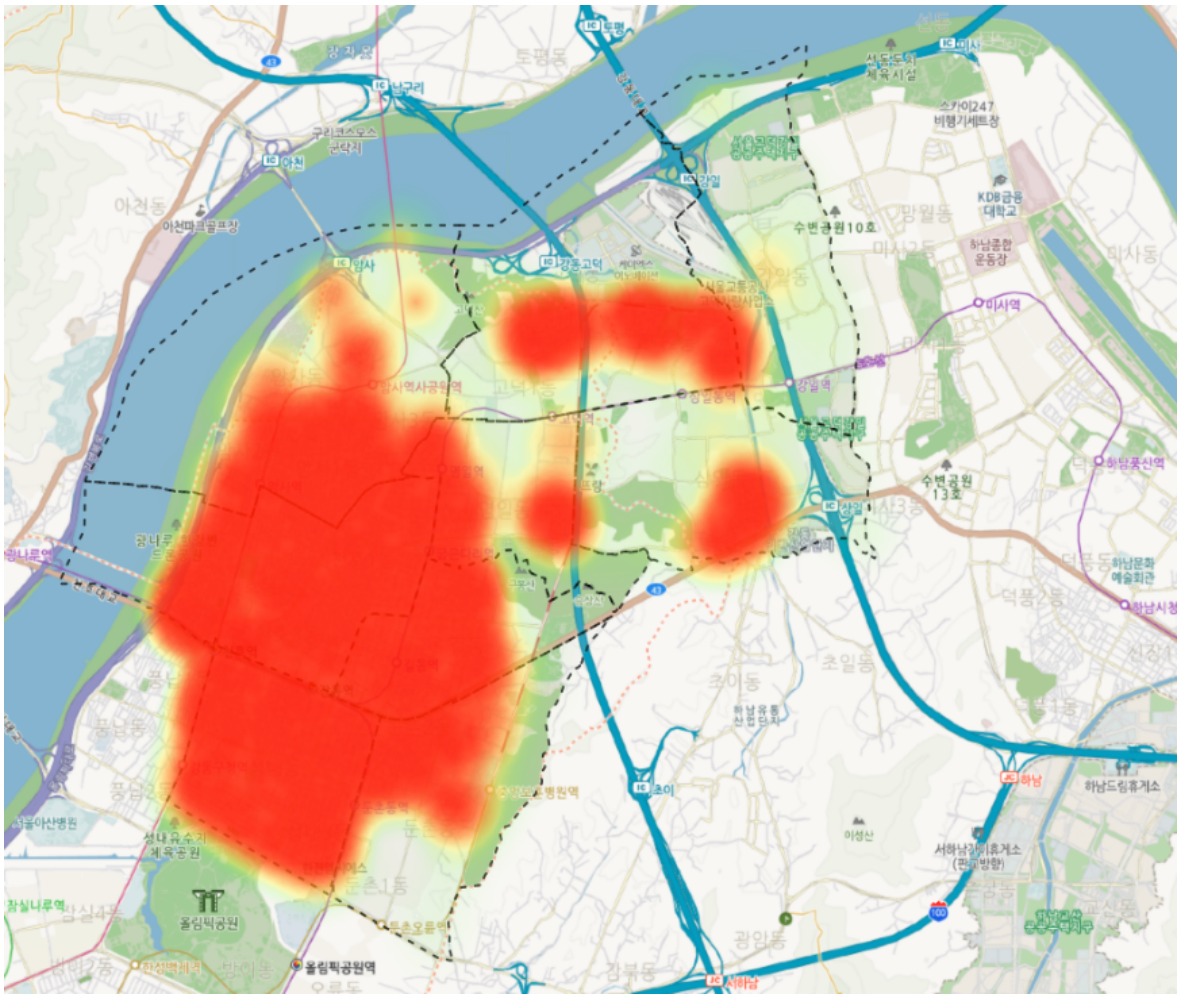

Figure 29: Spatial heat-demand intensity map for the Gangdong-gu district. Source: Seoul Energy Information System [38].